



ICM520: MACHINE LEARNING, AI & BIG DATA

ESG SCORE ANALYSIS



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Machine Learning Project: ESG Score Analysis

Component 1

1.1. Explanatory Data Analysis and Data Quality Assessment

The dataset consists of 2000 corporations' scores of their environmental, social and governance components. These features are predominantly numerical (continuous scores 0-10 or 0.0-1.0 and ordinal rankings 0-2, 1-5), with one category variable. Overall, we have 8 environmental variables, 4 social variables, and 3 governance variables. The “category” variable represents the classification of investment profile we aim to predict. Table 1 illustrates feature types to guide cleaning, scaling, and modelling of the data to ensure an interpretable result suitable for an ESG investment portfolio.

Table 1: Features, Data types and Missing Values in each column

Variable	Data Type	Feature Type	Missing values
FEAT_ENV1	int64	Ordinal (0-10 score)	0
FEAT_ENV2	float64	Numeric	0
FEAT_ENV3	int64	Ordinal (3-level scale)	0
FEAT_ENV4	int64	Ordinal (ranked levels)	0
FEAT_ENV5	int64	Ordinal (0-10 score)	0
FEAT_ENV6	float64	Numeric	0
FEAT_ENV7	int64	Ordinal (3-level scale)	0
FEAT_ENV8	int64	Ordinal (ranked levels)	0
FEAT_SOC1	int64	Ordinal (0-10 score)	0
FEAT_SOC2	float64	Numeric	0
FEAT_SOC3	float64	Numeric	0
FEAT_SOC4	float64	Negative numeric	0
FEAT_SOC5	float64	Numeric	0
FEAT_GOV1	str	Nominal Categorical	0
FEAT_GOV2	int64	Ordinal(0-10 score)	0
FEAT_GOV3	int32	Ordinal (0-10 score)	184
CATEGORY	Int64	Ordinal (ranked groups)	0

1.2. Data Processing and Cleaning

The dataset is complete with no missing values, except for FEAT_GOV3. The missing values were imputed using the median category. Although this approach inflates the central category and slightly reduces variance, it preserves its ordinality (Emmanuel et al, 2021). However, from a portfolio management perspective, this introduces a slight bias that firms with missing governance data were assigned ‘average’ characteristics.

Summary statistics in Table 2 highlight that most 0-10 scores centre around their midpoint with a tight spread, while continuous 0.0 – 1.0 scores display a moderate spread. Ordinal 1-5 and 0-2 rank level features also cluster around the central categories. The distribution has two key implications;

- The clustering of ESG scores risks firms appearing similar across ESG dimensions, reducing diversification opportunities (Bertelli and Torricelli, 2023).

- ii. It also reduces the discriminative power, limiting their effectiveness to distinguish top-performing firms (Berg Koelbel and Rigobon, 2022, 2022).

SOC4 spans from -1 to 0, which may measure the net harm, sentiment, or adverse social controversy impact spanning from “very severe to minimal”, which represents a company with legal risk (MSCI, 2020). Moreover, an important data quality issue is observed in governance feature 2, where an extreme outlier (value = 1000) lies outside the range 0 to 10. This anomaly is illustrated in Figure 1. This indicates a miscoded value rather than a meaningful score. Hence, the extreme maximum of 1000 is replaced by 10. This is important to the data quality check step, as extreme outliers can distort the segmentation of firms in modelling, giving misrepresented investment results.

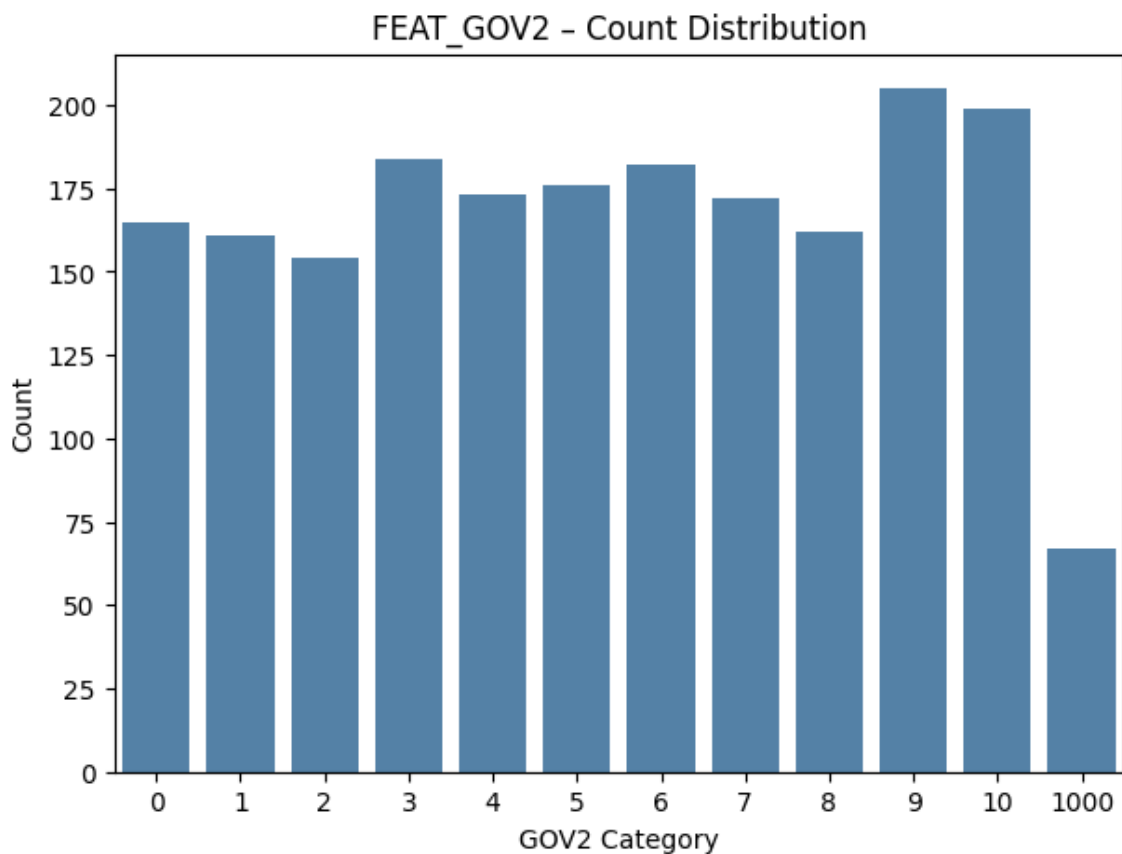


Figure 1: Bar Chart of Feature Governance 2 score distribution: Identification of Outlier

Table 2: Summary Statistics of Environmental, Social and Governance Features

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
FEAT_ENV1	2000	5.03	3.10	0	2	5	8	10
FEAT_ENV2	2000	0.33	0.30	0	0.06	0.25	0.56	1
FEAT_ENV3	2000	0.99	0.70	0	1	1	1	2
FEAT_ENV4	2000	3.04	1.20	1	2	3	4	5
FEAT_ENV5	2000	5.00	3.18	0	2	5	8	10
FEAT_ENV6	2000	0.34	0.31	0	0.06	0.24	0.59	1
FEAT_ENV7	2000	0.99	0.71	0	0	1	1	2
FEAT_ENV8	2000	3.02	1.24	1	2	3	4	5
FEAT_SOC1	2000	4.96	3.24	0	2	5	8	10
FEAT_SOC2	2000	0.50	0.29	0	0.30	0.50	0.70	1

FEAT_SOC3	2000	0.34	0.30	0	0.06	0.26	0.58	1
FEAT_SOC4	2000	-0.33	0.29	-1	-0.55	-0.25	-0.07	0
FEAT_SOC5	2000	0.33	0.29	0	0.07	0.25	0.55	1
FEAT_GOV2	2000	38.51	179.1	0	3	5	8	1000
FEAT_GOV3	1816	5.00	3.22	0	2	5	8	10

Table 3: After adjusting for missing values and outliers of the governance features

FEAT_GOV2	2000	5.34	3.24	0.0	3.00	5.00	8.00	10.0
FEAT_GOV3	2000	4.99	3.06	0.0	2.00	5.00	7.00	10.0

Moreover, governance1, originally a categorical string variable, is mapped onto an ordinal rank form (0-3) so the statistical modelling can process them. Figure 2 shows the distribution of the governance 1 feature.

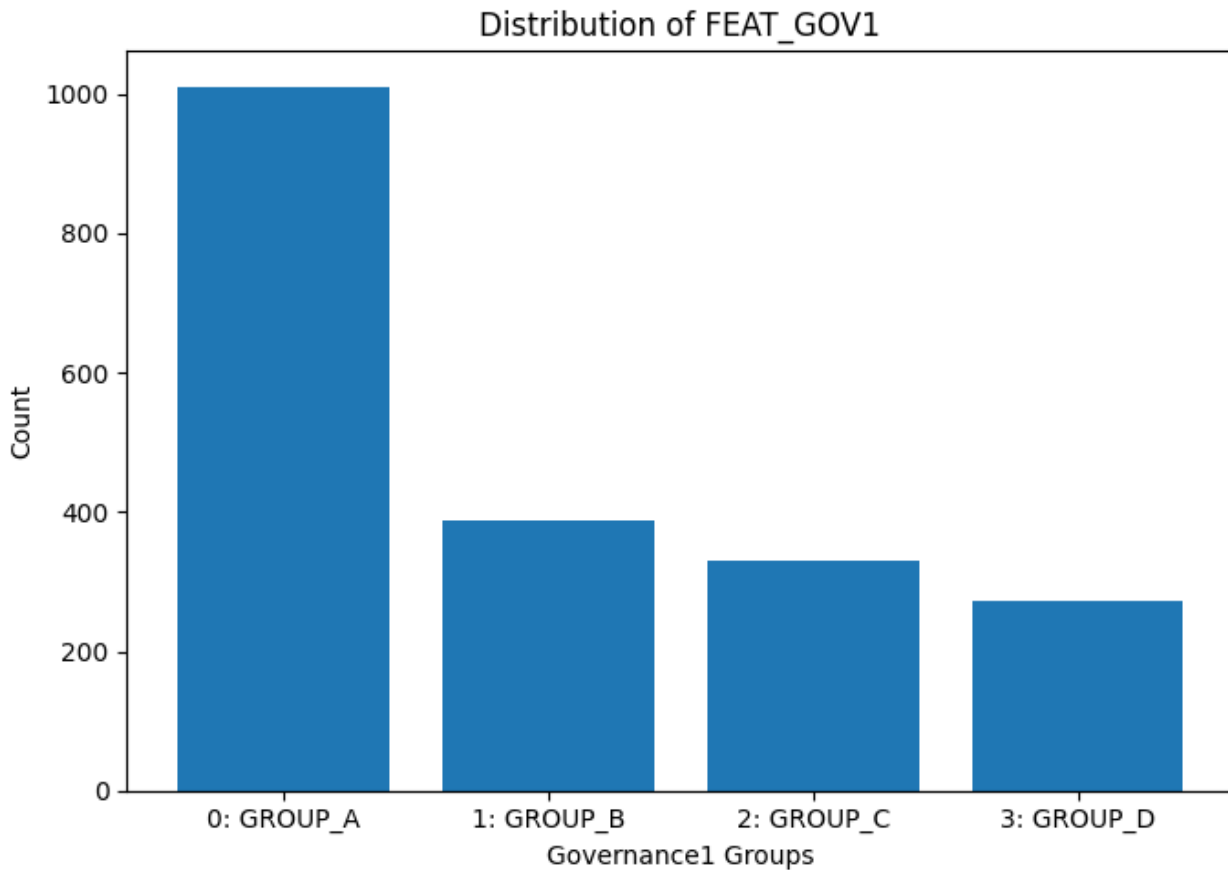


Figure 2: Bar Chart of Governance 1 Feature distribution

The data-processing step ensures that the inputs used for modelling are reliable and free from structural issues such as missing values and outliers, which can suppress underlying patterns and distort results during model fitting and undermine investment decisions.

1.3. Feature Distributions

Ordinal variables are visualised using count plots, while continuous numeric variables are visualised using histograms.

1.3.1. Environmental Features

The environmental features (FEAT_ENV1 and FEAT_ENV5) exhibit a very tight, uniform spread, which is characteristic of constructed ESG scores rather than raw environmental metrics (KnowESG, 2023). It is useful for ranking and categorising firms, but it also means these variables should be interpreted as high-level summary indicators rather than precise measures of underlying environmental risk.

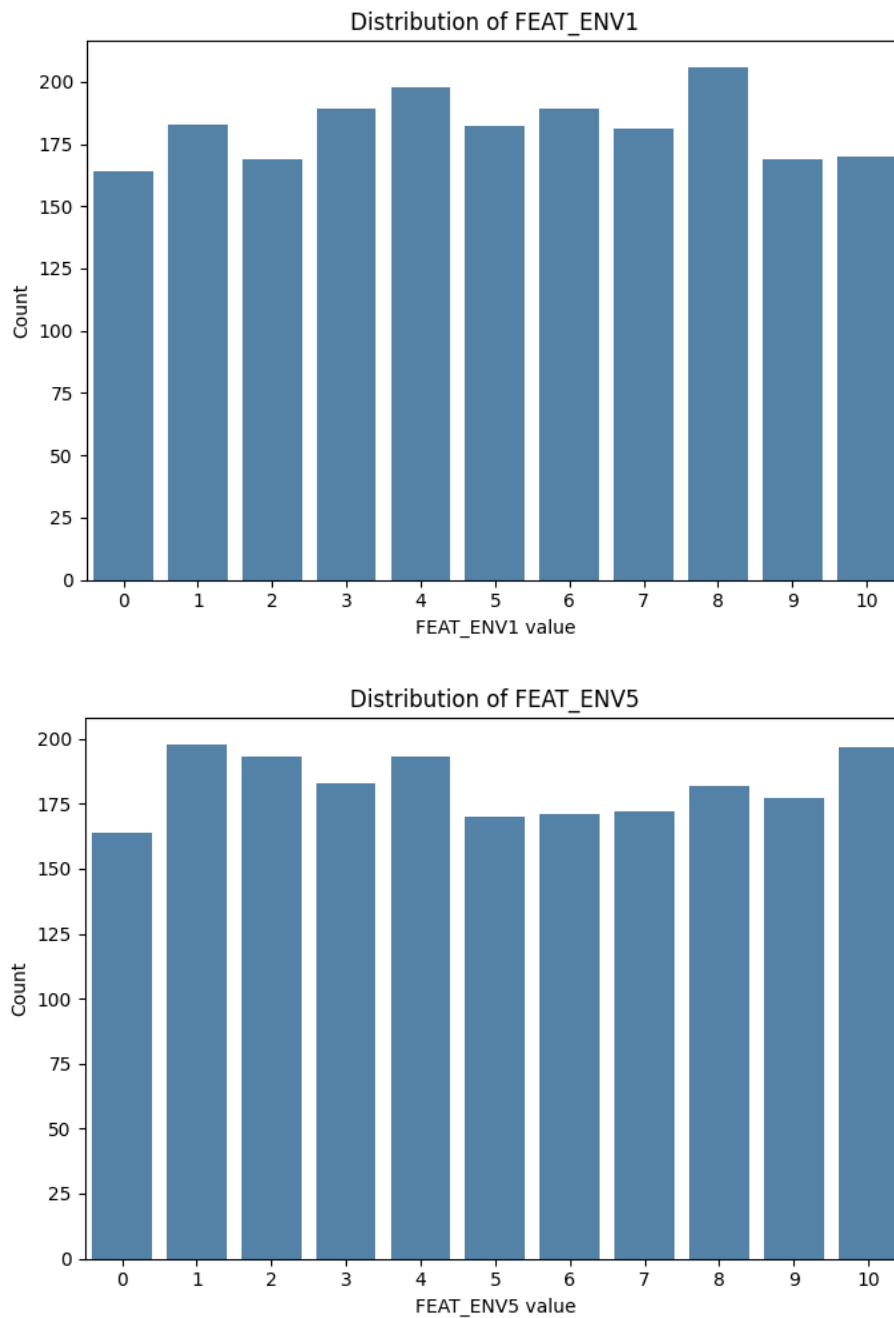


Figure 3: Count Plot of ENV1 and ENV5

The ordinal features (ENV3, ENV4, ENV7, ENV8) are highly concentrated around central categories. This pattern is consistent with the ESG rating providers, who apply conservative scoring methodologies and avoid assigning extreme values unless evidence is overwhelming (Berg, Koelbel and Rigobon, 2022). This central-tendency bias reduces variance, limits discriminatory power, and indicates that these ordinal features provide only “coarse-grained information” for predictive modelling (Berg, Koelbel and Rigobon, 2022). Interestingly, ordinal variables starting at 0 suggest ‘poor’ or ‘non-compliant’ represent the absence of positive performance, whereas 1-based scales implicitly assume a minimum level of compliance. Thus, these heterogeneous features require appropriate scaling before modelling and comparative analysis.

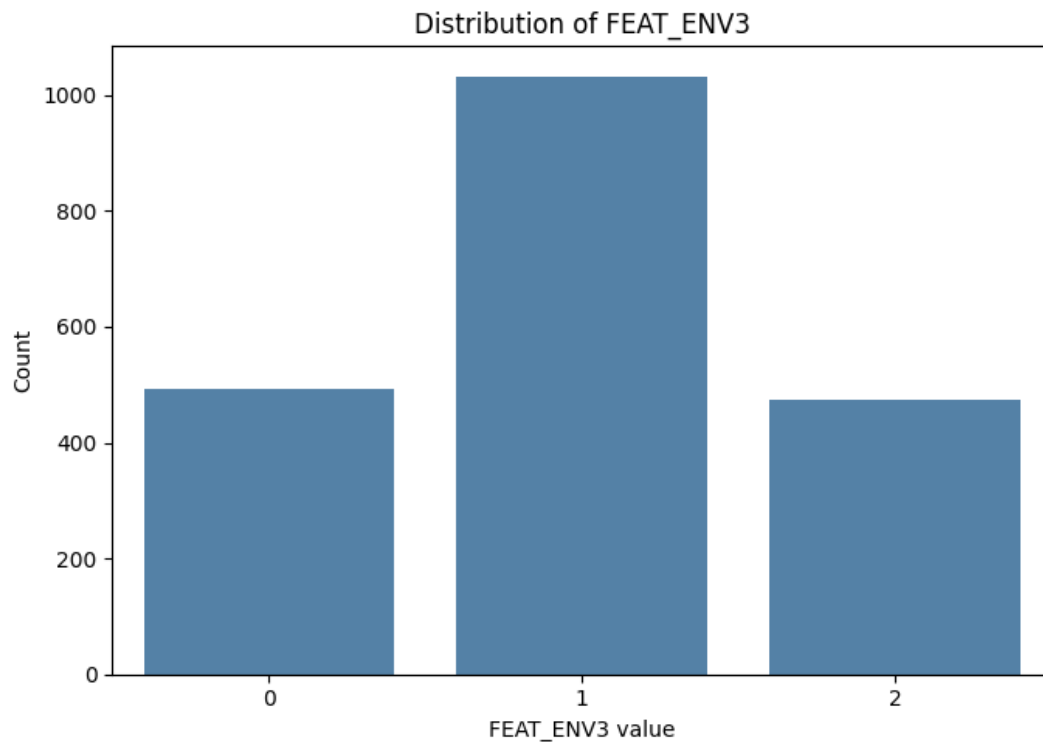


Figure 4: Ordinal Rank Feature Count Plot: ENV3

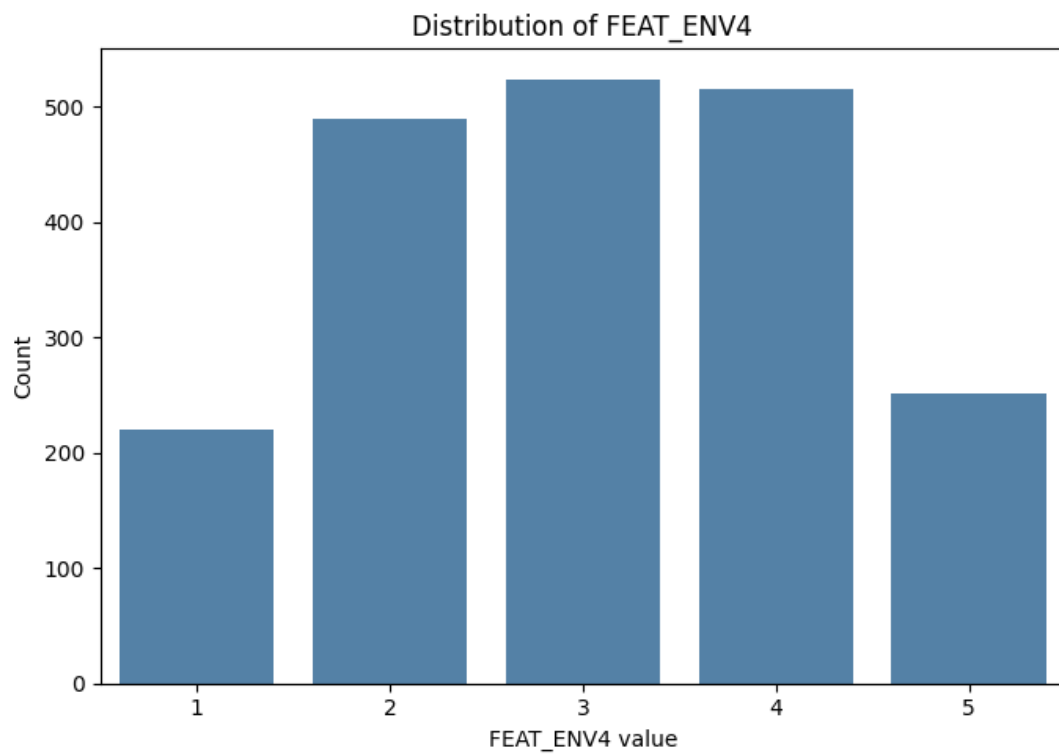
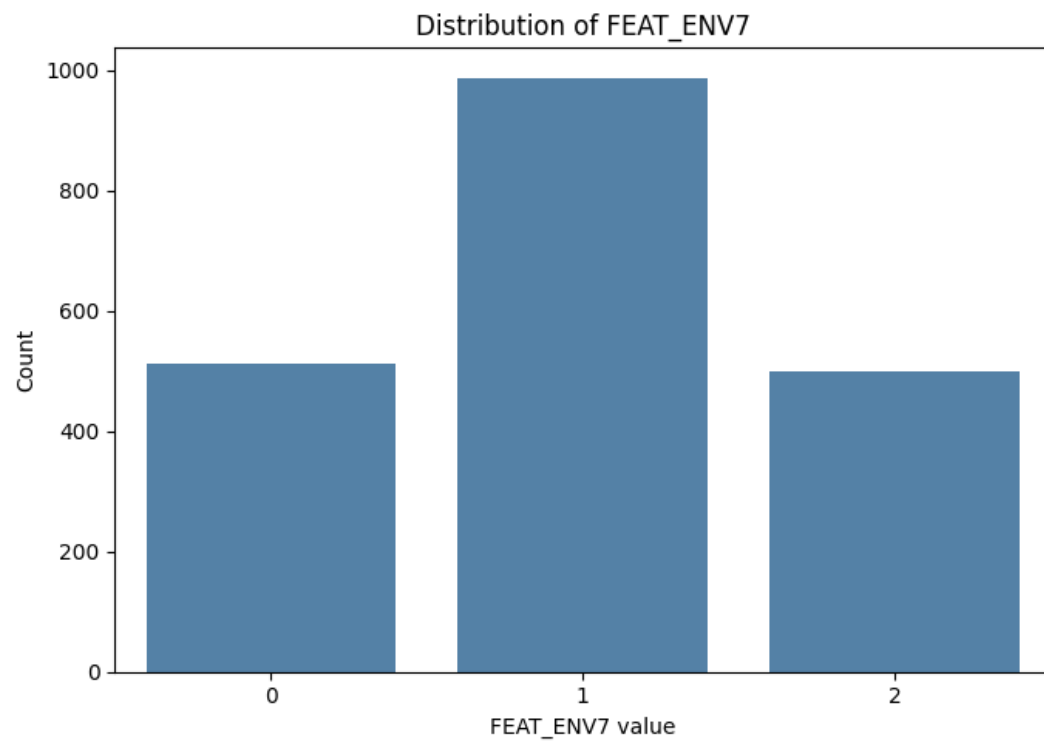


Figure 5: Ordinal Rank Features Count Plot: ENV4 and ENV7

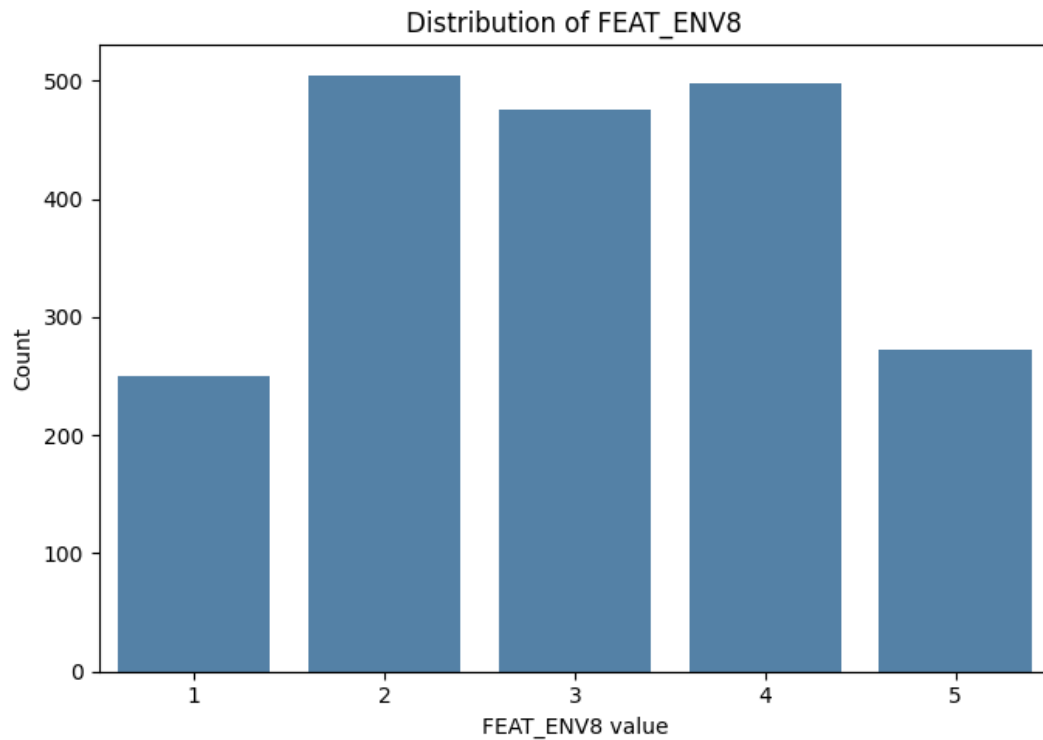


Figure 6: Ordinal Features Count Plot: FEAT_ENV8

Similarly, the continuous 0-1 environmental features (See Figure 7) exhibit a right-skewed distribution, typical of environmental proportion metrics, in which most firms demonstrate low levels of environmental achievement and only a few reach higher performance (Dorfleitner et al., 2020). Moreover, the near-identical distributions and low variance between the two variables may weaken their discriminatory power and increase redundancy. Low-variance variables with similar spread add limited information in segmenting firms or aligning the portfolio with sustainability investment preferences.

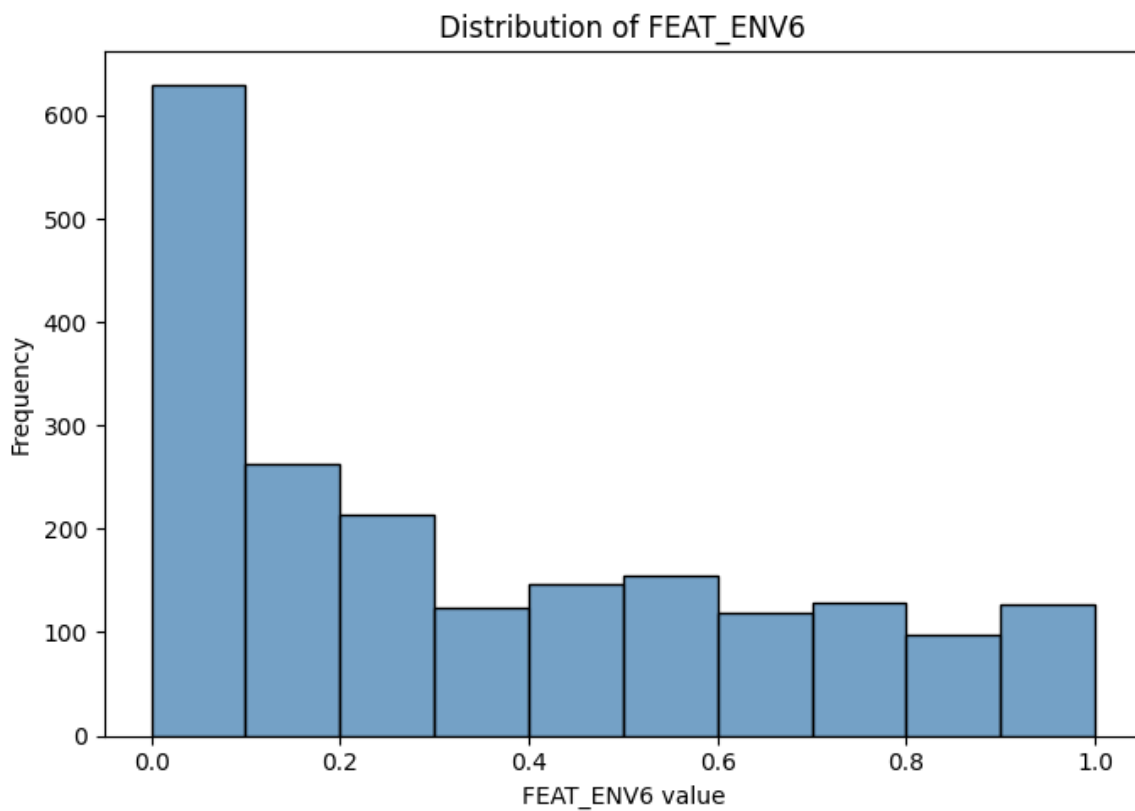
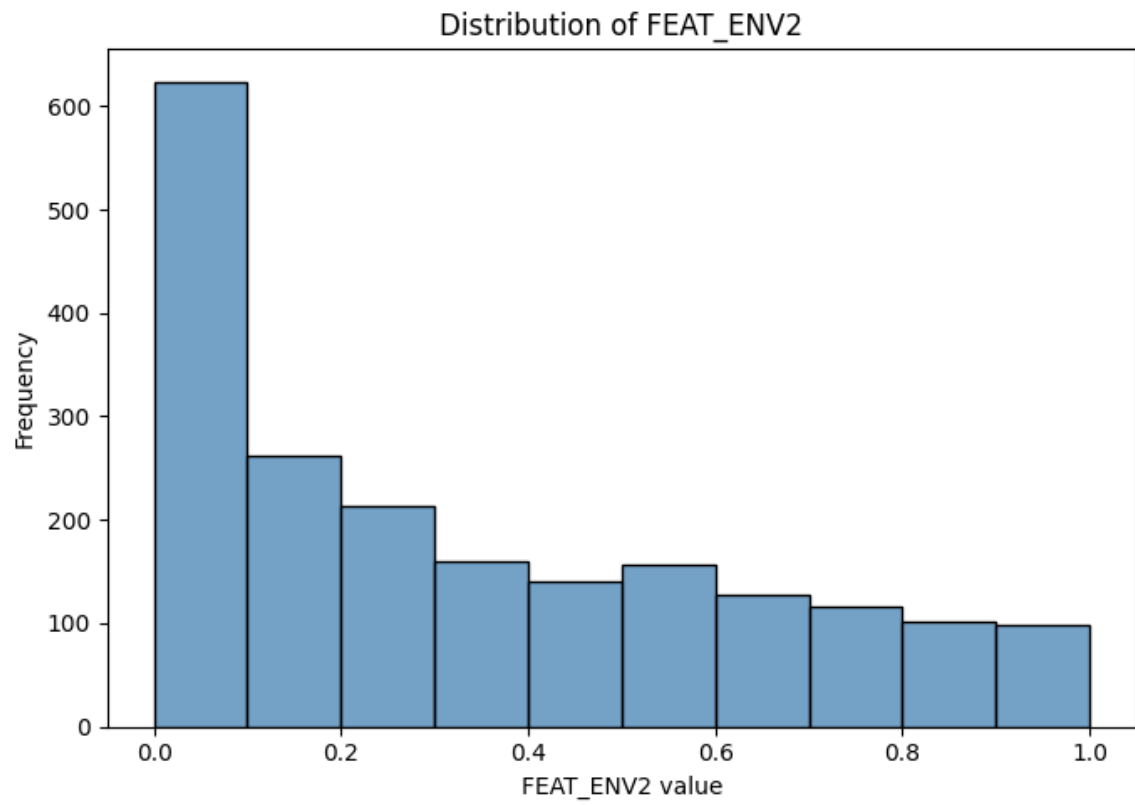


Figure 7: Histogram of Continuous Numeric ENV2 & ENV6 Features

1.3.2. Social Features

The social features SOC1 & SOC2 exhibit a similar distribution to environmental (ENV1 & 5) scores, with a tight spread and most firms rating concentrated around the mean and median. However, unlike SOC1, SOC2 exhibits a well-spread, with fewer observations at the boundaries, suggesting that the feature is not dominated by min/max values. There are fluctuations rather than concentration around the mean with peaks at 0.3, 0.7, and 0.9, illustrating that SOC2 gives better incremental information than SOC1 and the environment features.

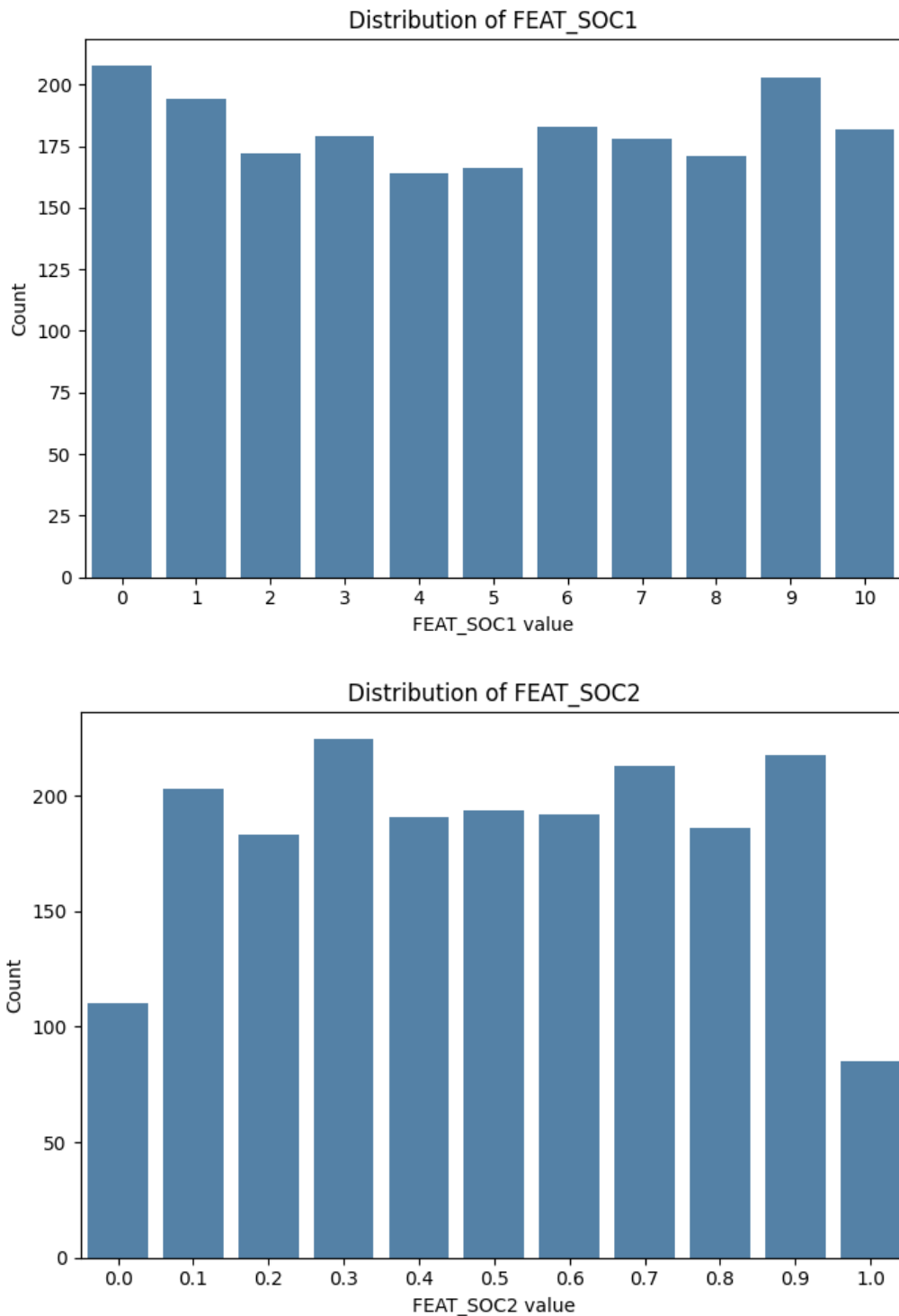


Figure 8: Ordinal Rank Feature Count Plot: SOC1 and SOC2

Numeric features (SOC3-5) concentrated around low scores show fewer firms reaching high scores of social performance. This reiterates that ESG rating providers apply conservative scoring methodologies and avoid assigning extreme values unless evidence is overwhelming (Berg, Koelbel and Rigobon, 2022). However, SOC4, the left-skewness explains less firms have minimal social controversy impact legal risk (MSCI, 2020).

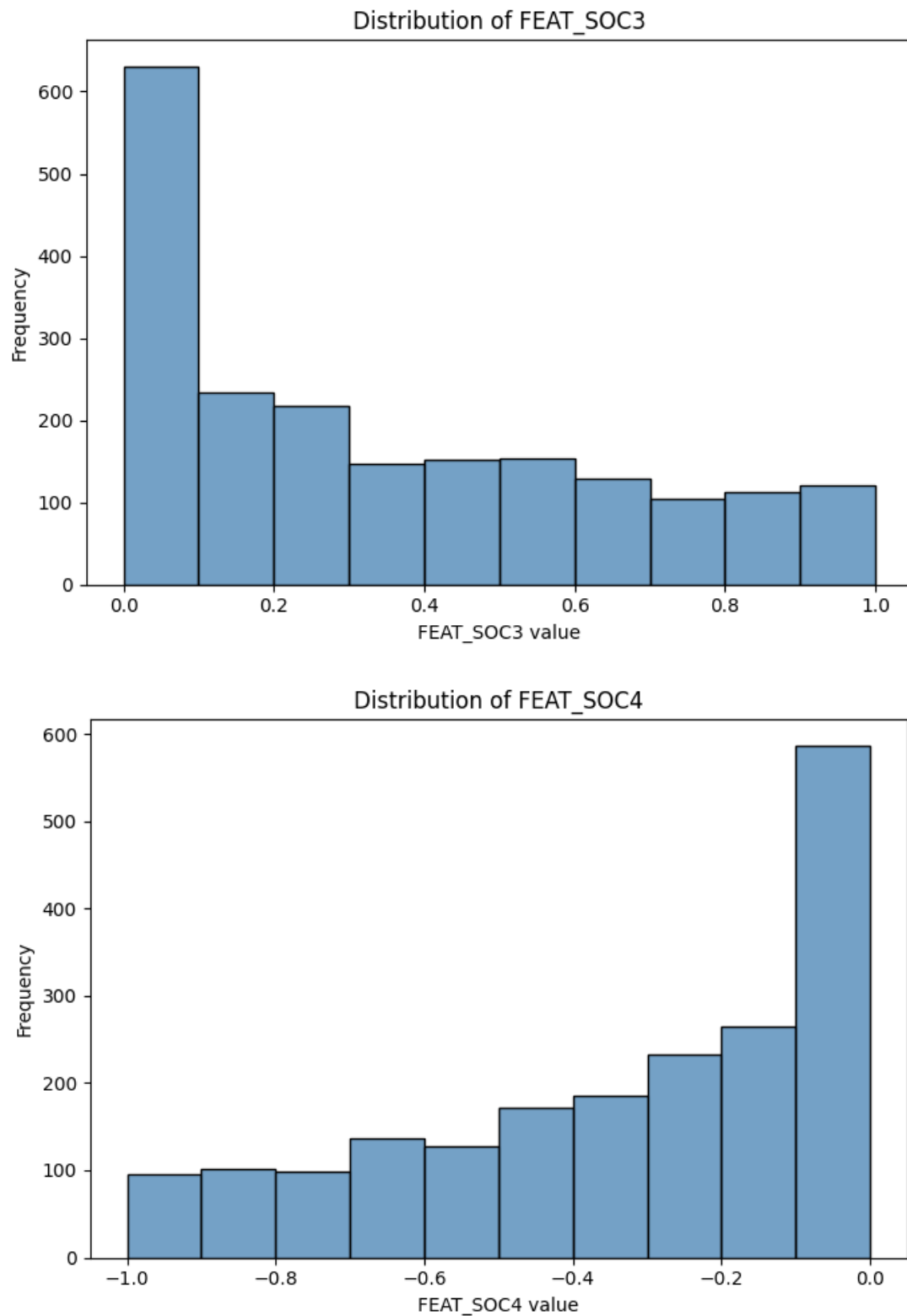


Figure 9: Histogram of Numeric Feature: SOC3 and SOC4

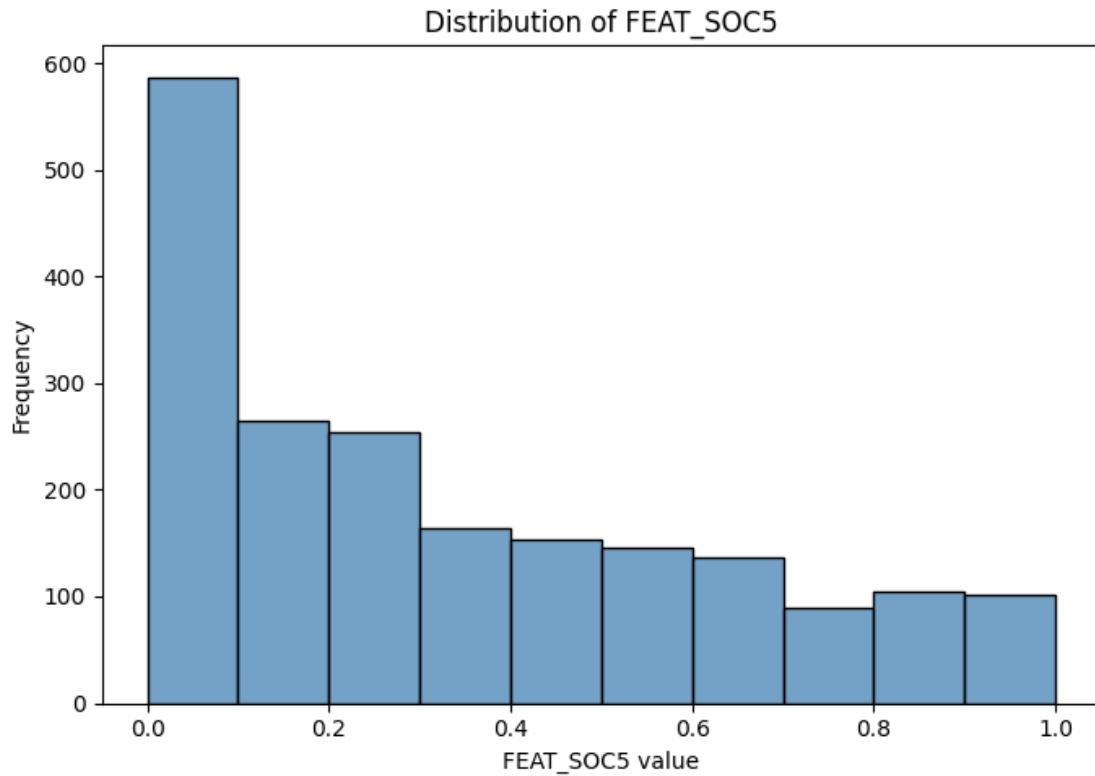


Figure 10: Histogram of Numeric Feature: SOC5

1.3.3. Governance Feature

The governance features show unbalanced structures due to outlier omission and missing value imputation. Figure 2 shows that GOV1 50% of the firms in Group A. The GOV3 missing values were replaced with the median, which has inflated the categorisation of the firms towards the median and slightly increased the spread. Moreover, replacing the 1000 with 10 meant that more firms fell in the latter category, which might be unrealistic as raters normally categorise firms within central limits with fewer concentration around 1 and 10 (Berg, Koelbel and Rigobon, 2022). From an investment portfolio perspective, the governance indicators should be interpreted cautiously, as they are not reliable in distinguishing firms for suitable governance-tilted portfolios. Moreover, they also contain weak discriminatory power for differentiating firms, making them weak predictors for the modelling.

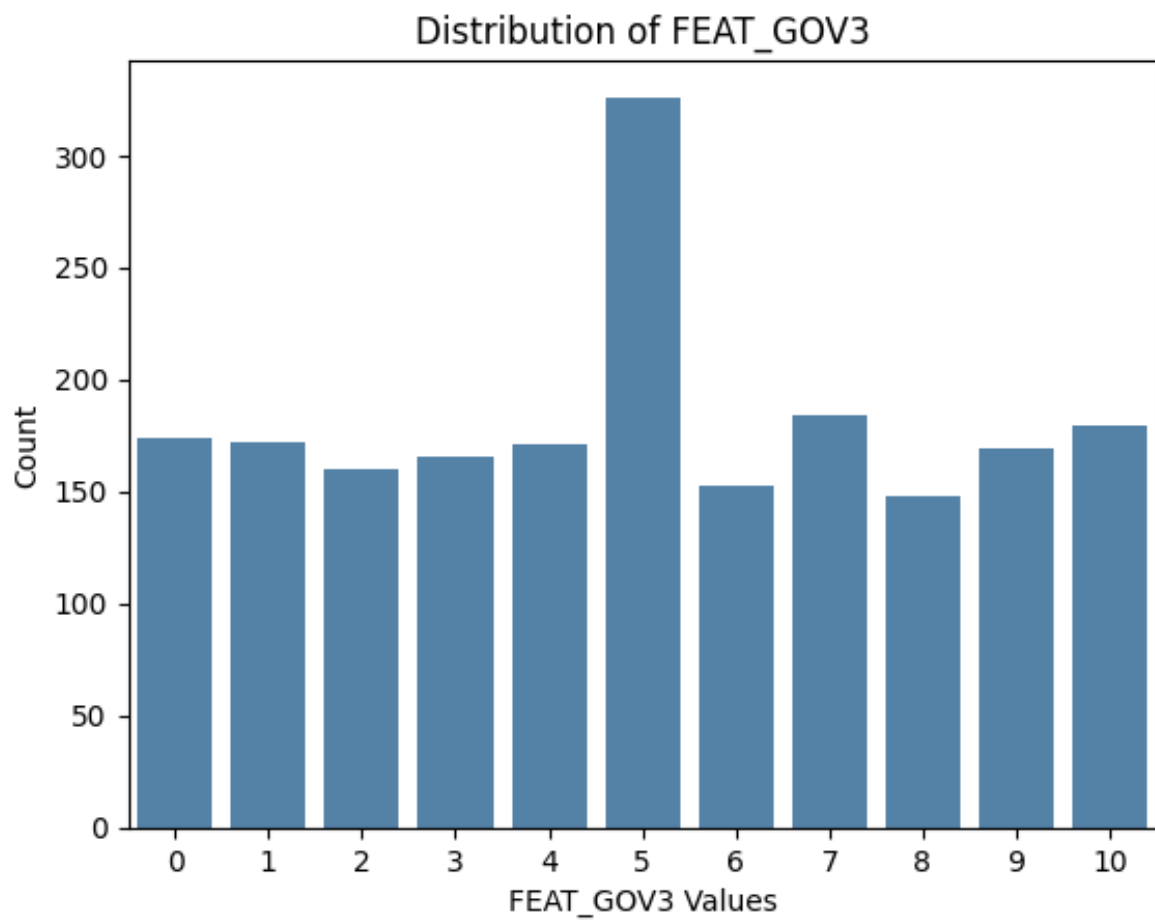
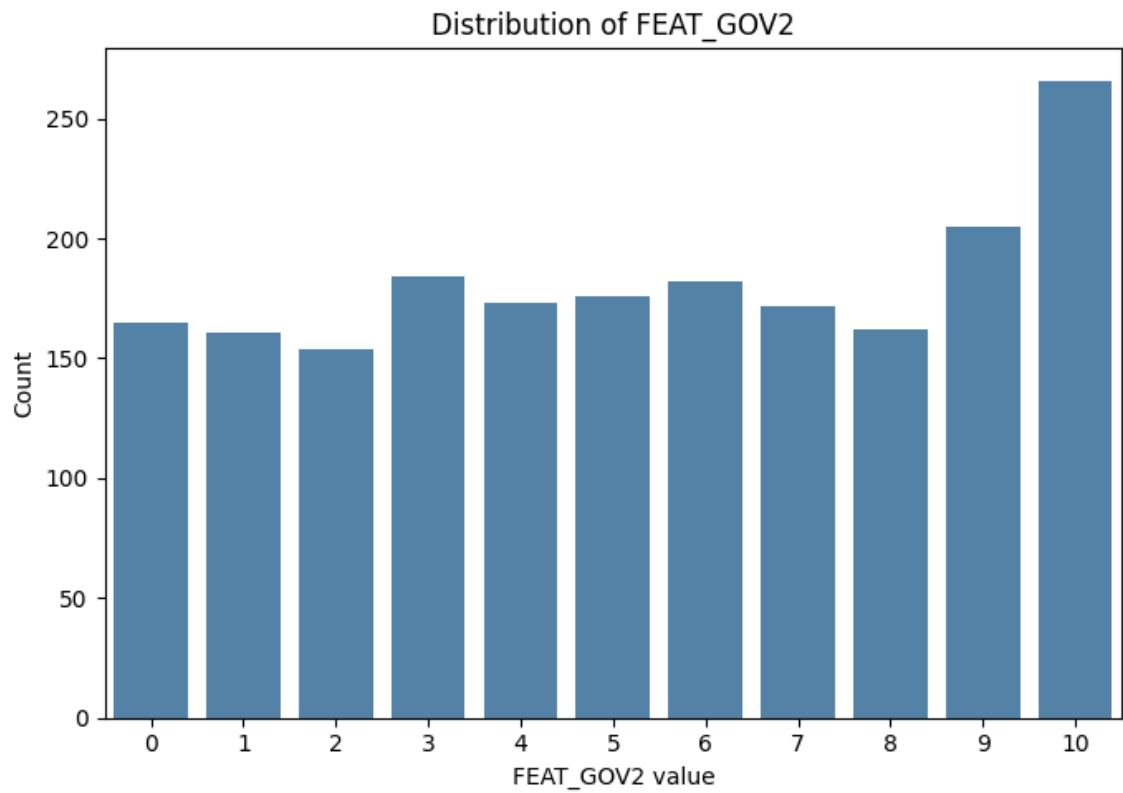


Figure 11: Ordinal Rank Feature Count Plot: GOV1 and GOV3

1.4. Correlation Matrix

The correlation analysis shows that ESG features operate largely independently. Cramer's V matrix is used for low-ordinal rank features (ENV3, ENV4, ENV8 and GOV1) and Pearson's correlation matrix for continuous and higher ordinal rank features (ENV1, ENV2, ENV5, ENV6, SOC3, SOC4, SOC5, GOV2, GOV3). Overall, there is a very weak association across the scores, suggesting each feature captures a distinct aspect of firm behaviours, even though most features showed similar distributions, but there is no strong overlapping information evidence. For an investment portfolio perspective, this means that the dataset provides granular information for assessing firms for sustainability preferences. However, the perfect negative correlation between SOC4 and SOC5 highlights that if one increases, the other feature decreases in the same predictable way, suggesting perfect inverse behaviour. SOC4 is removed from the feature set to ensure that remaining social metrics contribute distinct, non-conflicting insights when profiling firms against sustainability preferences.

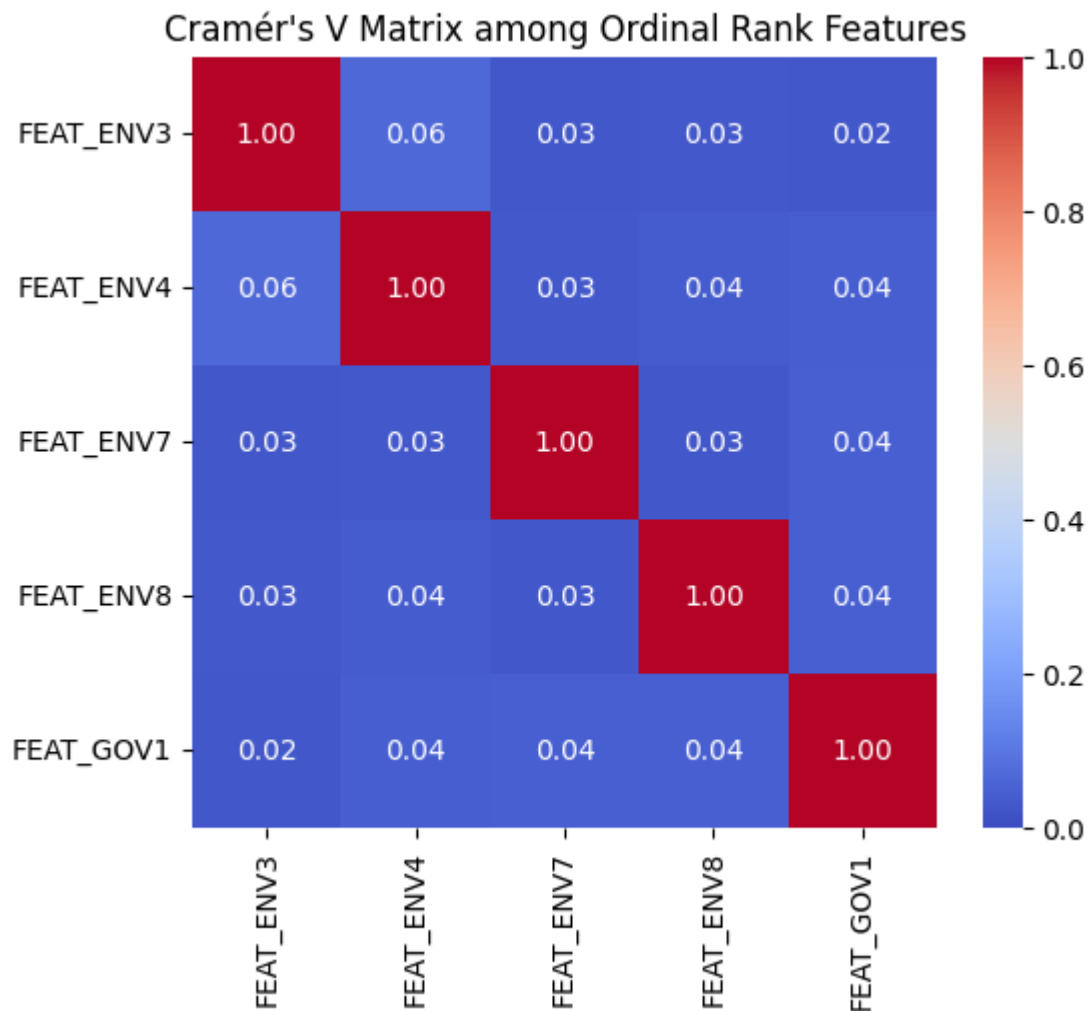


Figure 12: Crammer's V Matrix for Low-Ordinal Rank Feature with (0-2, 0-3 and 1-5 ranking levels): ENV3, ENV4, ENV8 and GOV1

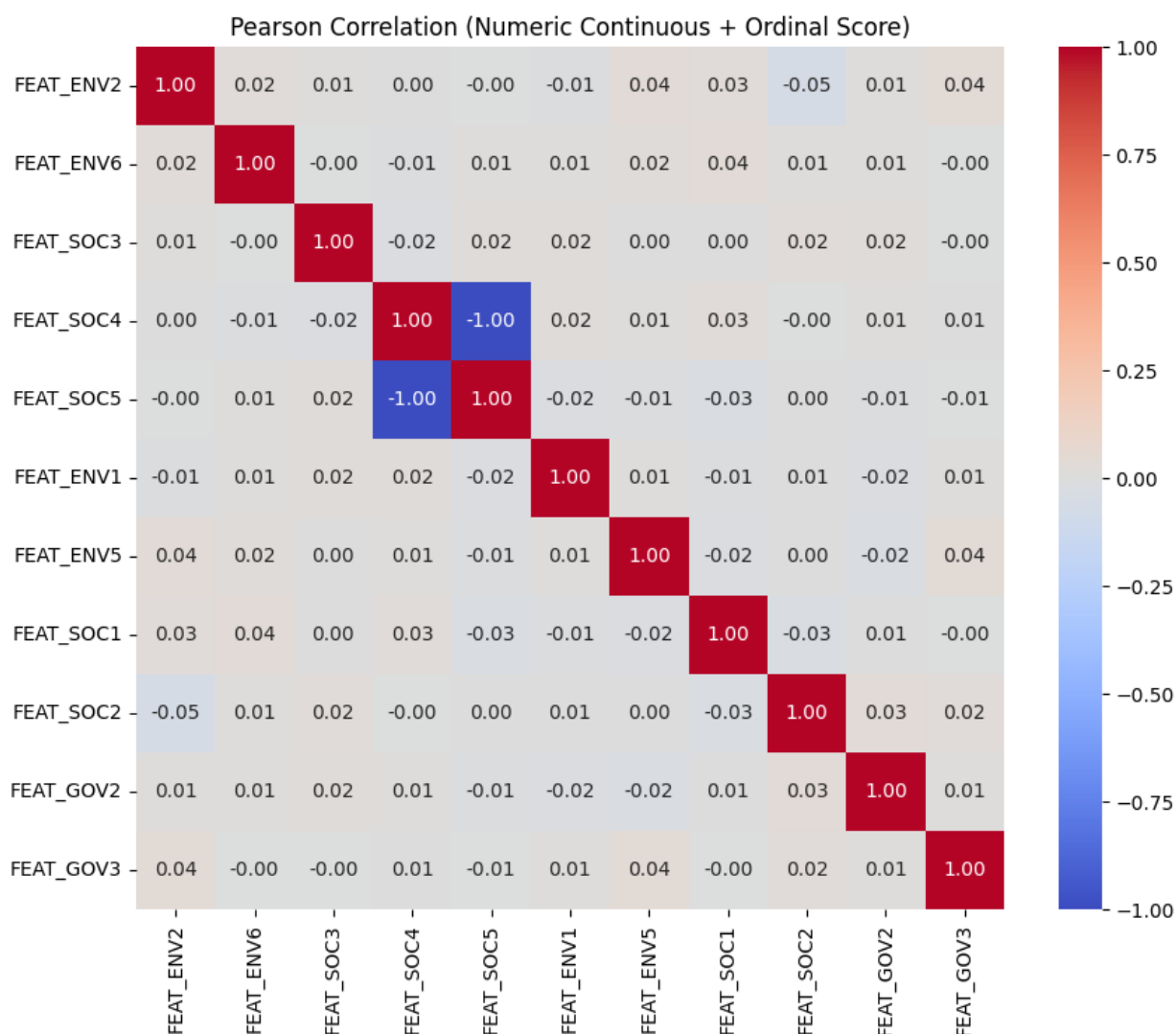


Figure 13: Pearson Correlation for Numeric and High Ordinality Features (0-10 ranks): ENV1, ENV2, ENV5, ENV6, SOC3, SOC4, SOC5, GOV2, GOV3

1.5. Data Leakage

The correlation analysis between ESG features and the CATEGORY variable provides insight into the predictive relevance of each feature. SOC1 shows the strongest association (0.46), indicating that it carries meaningful predictive information but also poses the highest risk of embedding outcome-related patterns directly into the model. High correlation with the target variable implies data feature usefulness for prediction. Furthermore, SOC2 (0.33) and SOC3 (0.28) show moderate relationships, suggesting they contribute useful behavioural or social-risk signals without dominating the classification structure. Interesting to note that, as section 1.1.2 indicated, SOC2 has a more balanced distribution than SOC1, but is moderately associated with the Category variable. Moreover, environmental features and governance features have weak or very weak (<0.07) associations with CATEGORY, meaning they contribute little predictive value and are unlikely to influence portfolio decisions. From an investment portfolio perspective, only a small subset of features meaningfully aligns with firm-categorising, while the remainder function as contextual descriptors of firms' ESG outlook. Overall, there is minimal data leakage risk in the dataset.

Table 4: Correlation Analysis of ESG features with Category Variable

Correlation Analysis with the CATEGORY variable	
Crammer's V for Low Cardinal features & Personas' Correlation for High cardinality (0-10 scores) and numeric features	
Feature	Corelation
FEAT_SOC1	0.46
FEAT_SOC2	0.33
FEAT_SOC3	0.28
FEAT_SOC4, FEAT_SOC5	0.28
FEAT_ENV7	0.23
FEAT_ENV5	0.21
FEAT_ENV8	0.19
FEAT_ENV2	0.07
FEAT_GOV3	0.05
FEAT_ENV4	0.04
FEAT_ENV3	0.03
FEAT_ENV6	0.02
FEAT_GOV1	0.02
FEAT_ENV1, FEAT_GOV2	<0.01

1.6. Live-Trading Strategy

Category labels are produced through subject-matter expert judgement without a documented scoring methodology, potentially reflecting embedded bias and inconsistency rather than objective ESG scoring. In a live trading environment, the absence of a temporal dimension may lead to stale classifications and reputational risk as the model cannot detect the over-time change in corporations' ESG behaviour. The gap between stated ESG commitments and actual performance is only visible if the company is tracked across time, enabling Greenwashing (Kathan et al., 2025). From an investment portfolio, operating in live markets, relying on such non-transparent, non-temporal ESG scores may impair suitability assessments, distort portfolio construction, and expose to misalignment between expected and realised sustainability outcomes. The class distribution Figure 14 shows a clear imbalance across the four categories, with Category 1: ESG Normal, dominating the dataset and Categories 0 and 2 forming smaller segments about 20.7% and 18.7%, respectively, while Category 3: ESG Champion is extremely underrepresented (2.4%). For investment-profiling and live-trading applications, the imbalance means the model will learn more about the firms in the majority class, producing more stable predictions but weaker, less reliable classifications for rare categories. Misclassification reduces the model's ability to identify genuinely atypical or high-risk firms, requiring greater oversight when interpreting minority-class signals like ESG Champion firms to prevent errors.

Table 5: Firms Category classification and percentage

CATEGORY	Percentage
0: ESG Loser	20.65
1: ESG Normal	58.30
2: ESG Winner	18.65
3: ESG Champion	2.40

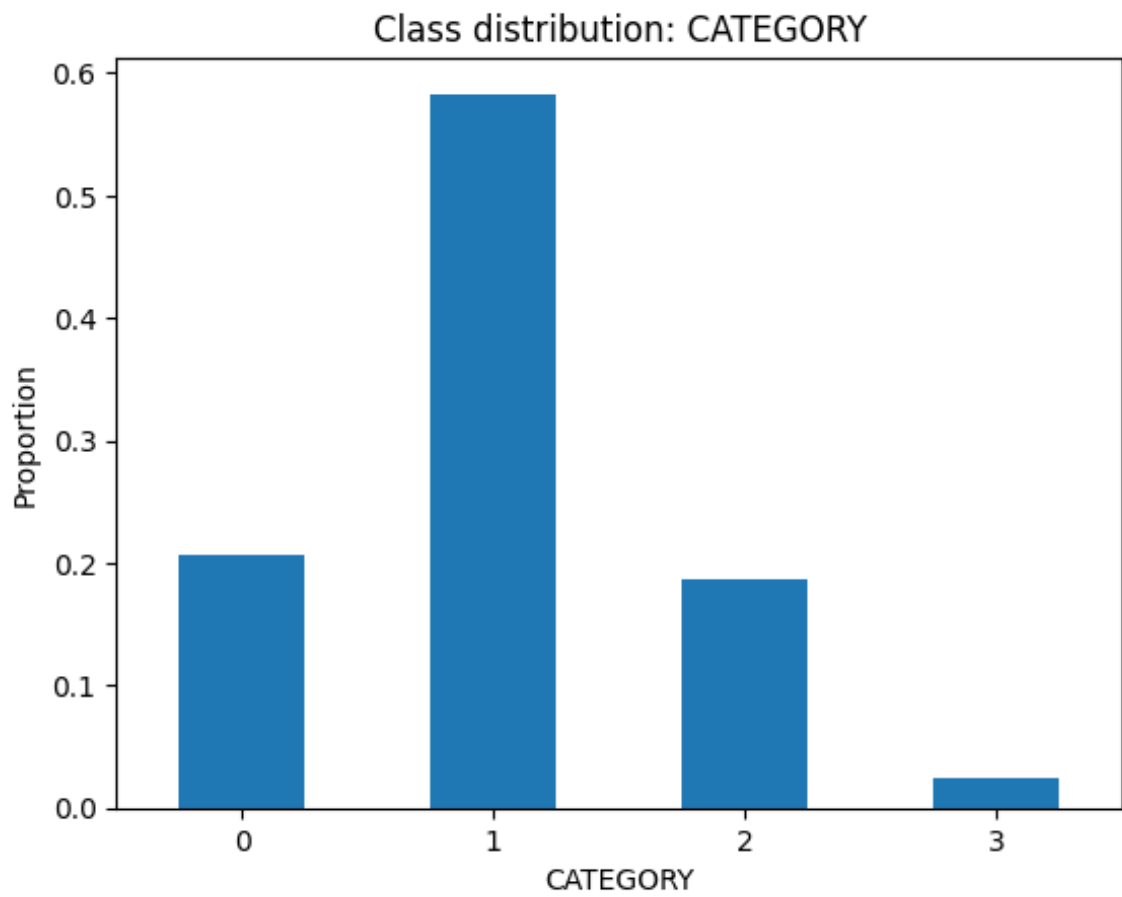


Figure 14: Proportion Bar Chart of Category: Classifications of the Firms

Component 2

This component evaluates four classification models with the objectives of predictive accuracy, model stability, explainability, client reporting and governance compliance. The model uses the 2000 corporation as a training set and 1000 new firms as a test dataset. The four models built are;

- i. Multinomial Logistic Regression (baseline model)
- ii. Elastic Net Logistic Regression
- iii. Gradient Boosting
- iv. Random Forest

2.1. Feature Engineering and Preprocessing

Firstly, before training the models, the data quality issues in the test set were addressed to ensure consistency. It included imputing missing values for GOV3 with the median, handling outlier observations in GOV2, and encoding GOV1 as 0-3. Furthermore, the redundant feature SOC4 was dropped from the test set following the analysis in sections 1.4 and 1.5. Secondly, we separate the X (input) and y (output) in the training and test set. The input variables are the ESG feature set, and the output is the Category variable. Lastly, X_{train} and X_{test} sets are scaled using the Standard Scaler to ensure the variation between the low-cardinality, high-cardinality and numeric features is minimised. This step helps ensure each feature makes a fair contribution to explaining the variance in the target variable. Standard Scaler transforms the feature using

$$X_{Scaled} = \frac{X - \mu}{\sigma}$$

This transforms the feature set to have a mean of 0 and a standard deviation (spread) of 1 (GeeksforGeeks, 2020). This allows the models to focus on genuine relationships rather than differences in measurement units. However, the target category variable; y_{train} and y_{test} are not transformed as it represents the actual investment profiling category which models are trying to predict. In other words, we standardise the inputs to improve model learning, but keep the target exactly as defined, so the model's predictions remain aligned with the classification framework. Important to note is that the scaled features are used in Logistic Regressions, tree-based models, Gradient Boosting, and Random Forest, do not require scaling.

2.2. Model Deployment

2.2.1. Multinomial Logistic Regression

Logistic Regression is the baseline model due to its transparency, stability and long-term established use in financial risk modelling and client-profiling tasks. This model provides interpretable coefficients to understand how each ESG feature influences the probability of a firm belonging to a given Category (Altman and Barboza, 2024), supporting FCA and MiFID II requirements for model explainability (FCA – REF). Furthermore, Altman and Barboza (2024) also note that Logistics Regression requires fewer assumptions, reducing operational-risk exposure, and has demonstrated stable performance across a longer forecasting horizon. From an investment perspective, interpretability is crucial to assess why a model assigns a firm to a particular ESG category, especially when classification influences capital allocation.

The confusion matrix illustrates that Logistic regression performs consistently across the training and test sets. Misclassification mainly occurs between ESG Normal, Winner and Champion,

indicating boundary ambiguity rather than model failure. This means the model slightly overweight firms that appear marginally better, or delays recognising improving firms. The ROC curves reinforce this stability, with all classes achieving high AUC values, confirming strong ranking ability. The economic impact of misclassification between Loser and Normal is

$$Test\ Error\ Rate_{0 \rightarrow 1} = \frac{20}{198 + 20} = 9.17\%$$

This is low but meaningful misalignment, but not extreme, as classifying ‘Loser as Champion’.

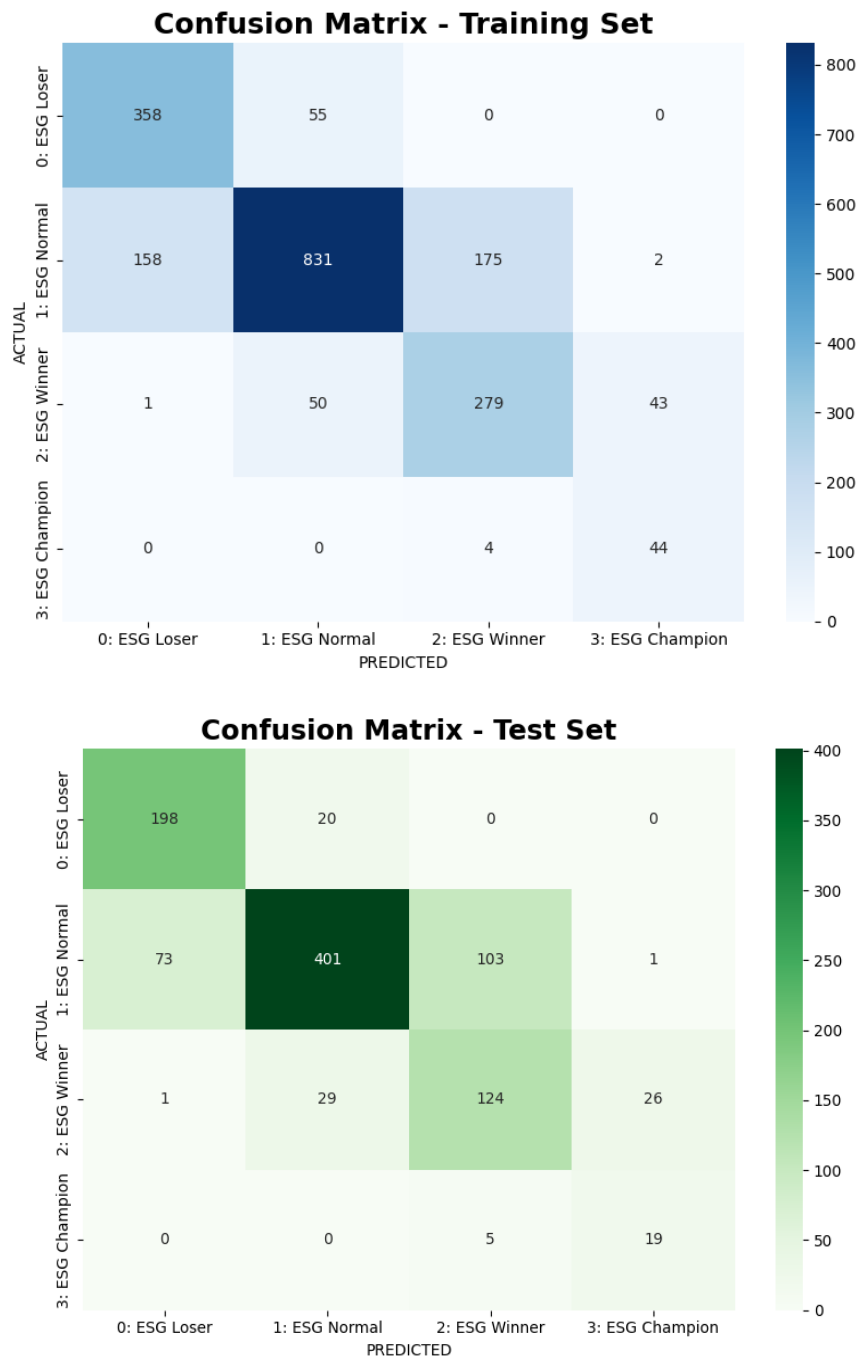


Figure 15: Confusion Matrix from Train and Test Set: Logistic Regression – Baseline Model

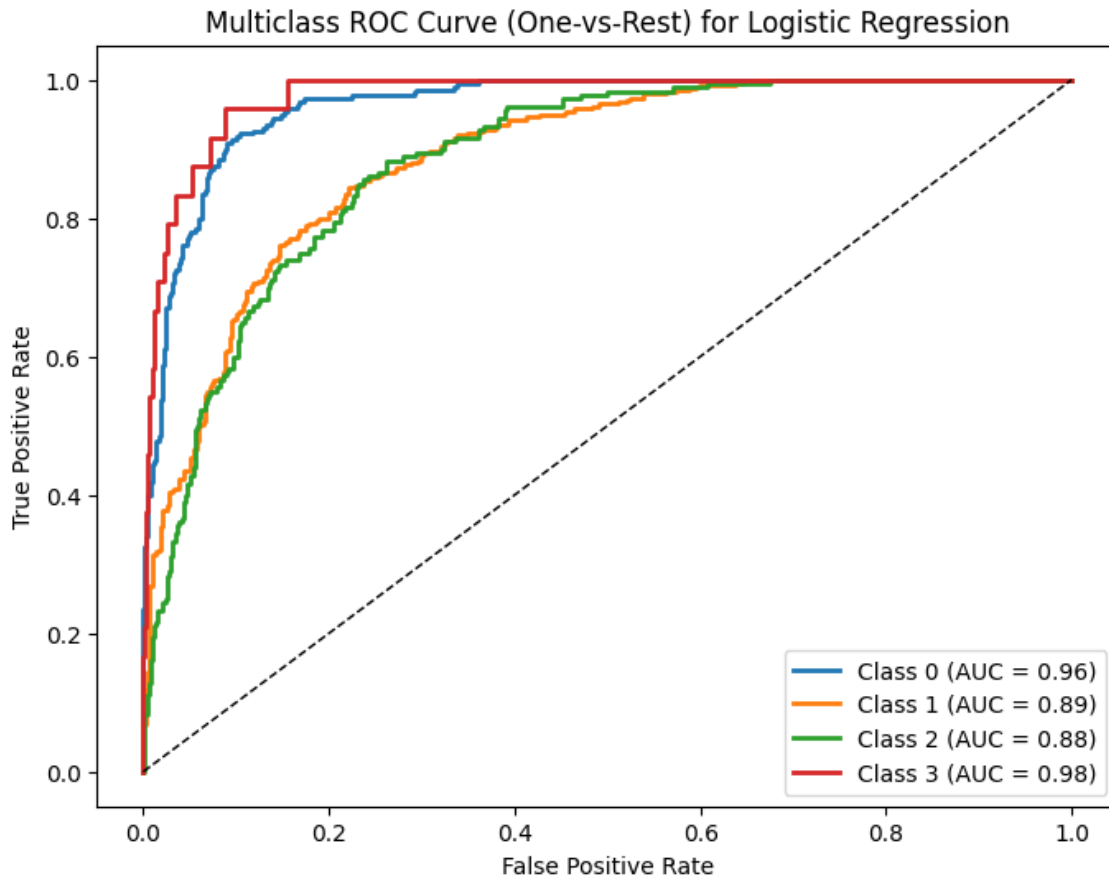


Figure 16: ROC curve: Logistic Regression – Baseline Model

Figure 17 shows that LR relies heavily on FEAT_SOC1 and FEAT_ENV7, dominating in importance, showing that ESG performance is primarily driven by social and environmental scores. This is also indicative of the correlation analysis in section 1.5.

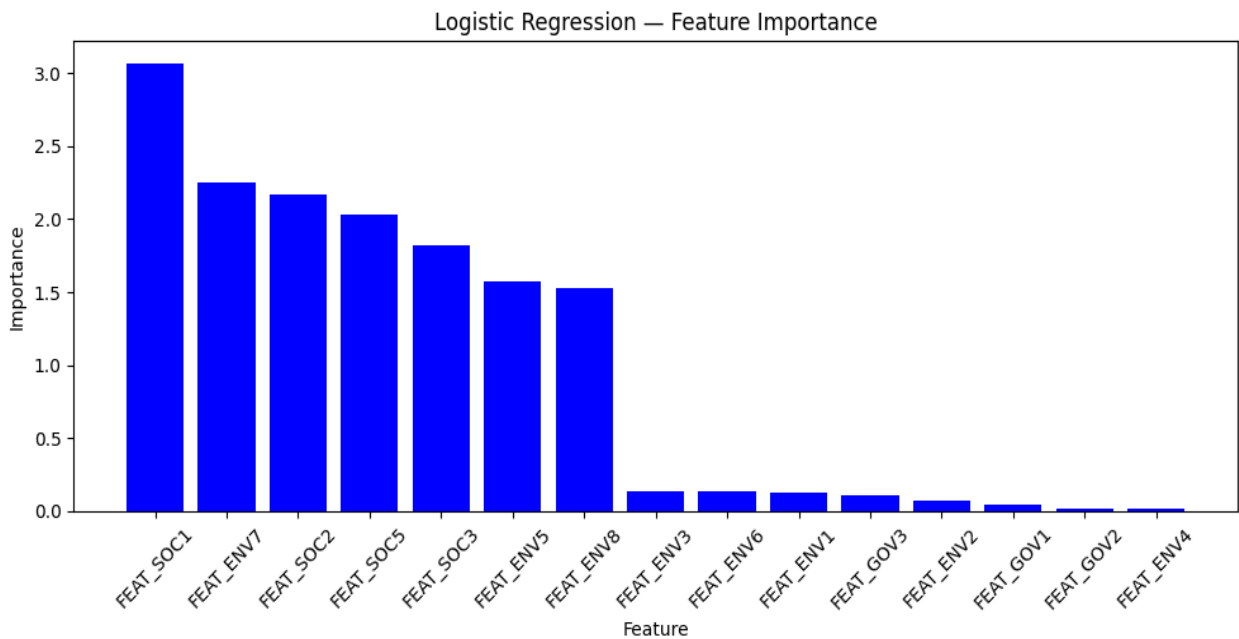


Figure 17: Feature Importance Plot: Logistic Regression

2.2.2. Elastic Net Logistic regression

Elastic net logistic model focuses on the most useful information when there are a lot of variables. It combines Lasso (L1) and Ridge penalties (L2), meaning one removes unnecessary features and the other smooths and stabilises the remaining ones, resulting in a model that is simple, more reliable and better at generalising to new data (Gogineni et al., 2026). From an investment portfolio perspective, this model gives stable and transparent ESG predictors effective for governance and regulatory purposes. The model performed well, with most predictions falling within the correct or adjacent ESG categories, with no severe misclassifications, and high AUC scores across all classes (0.89-0.98). However,

$$Test\ Error\ Rate_{0 \rightarrow 1} = \frac{58}{160 + 58} = 26.6\%$$

shows a meaningful classification error because the model upgrades a weak-performing firm into the mid-tier category, creating unintended exposure to high-risk, low-sustainable firms.

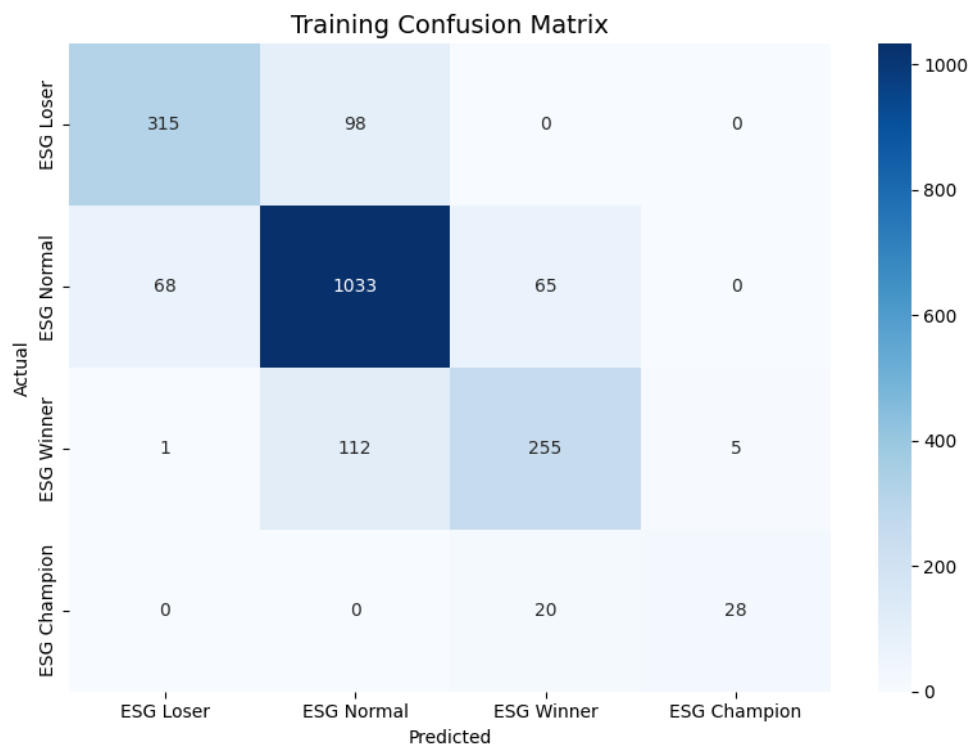


Figure 18: Confusion Matrix on Training Set: Elastic Net Logistic Regression

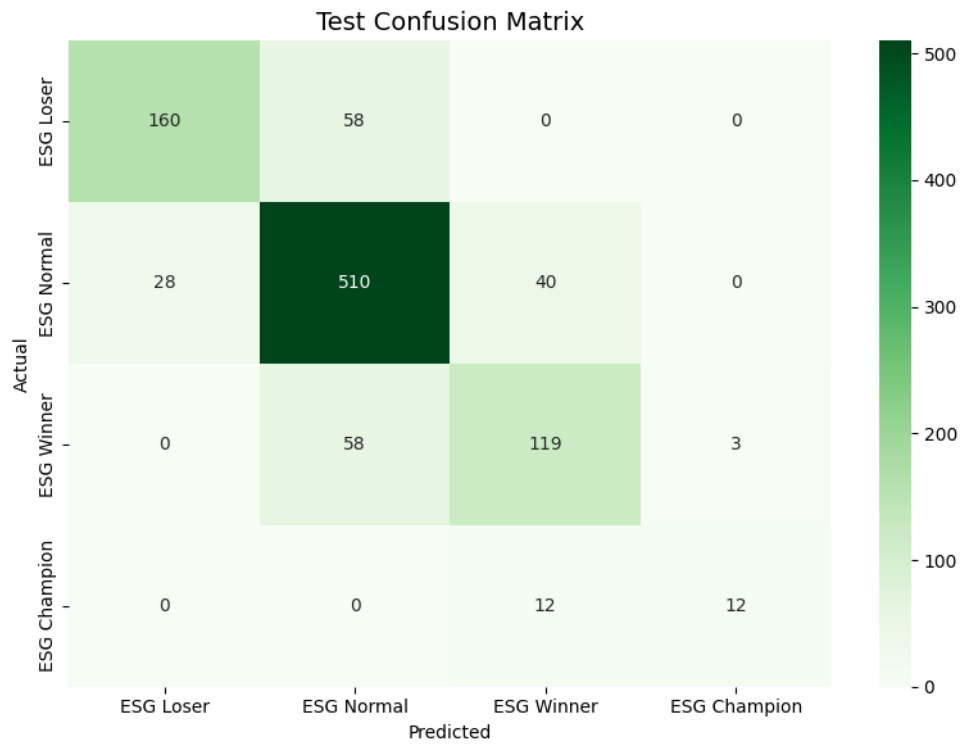


Figure 19: Confusion Matrices on Test Set: Elastic Net Logistic Regression

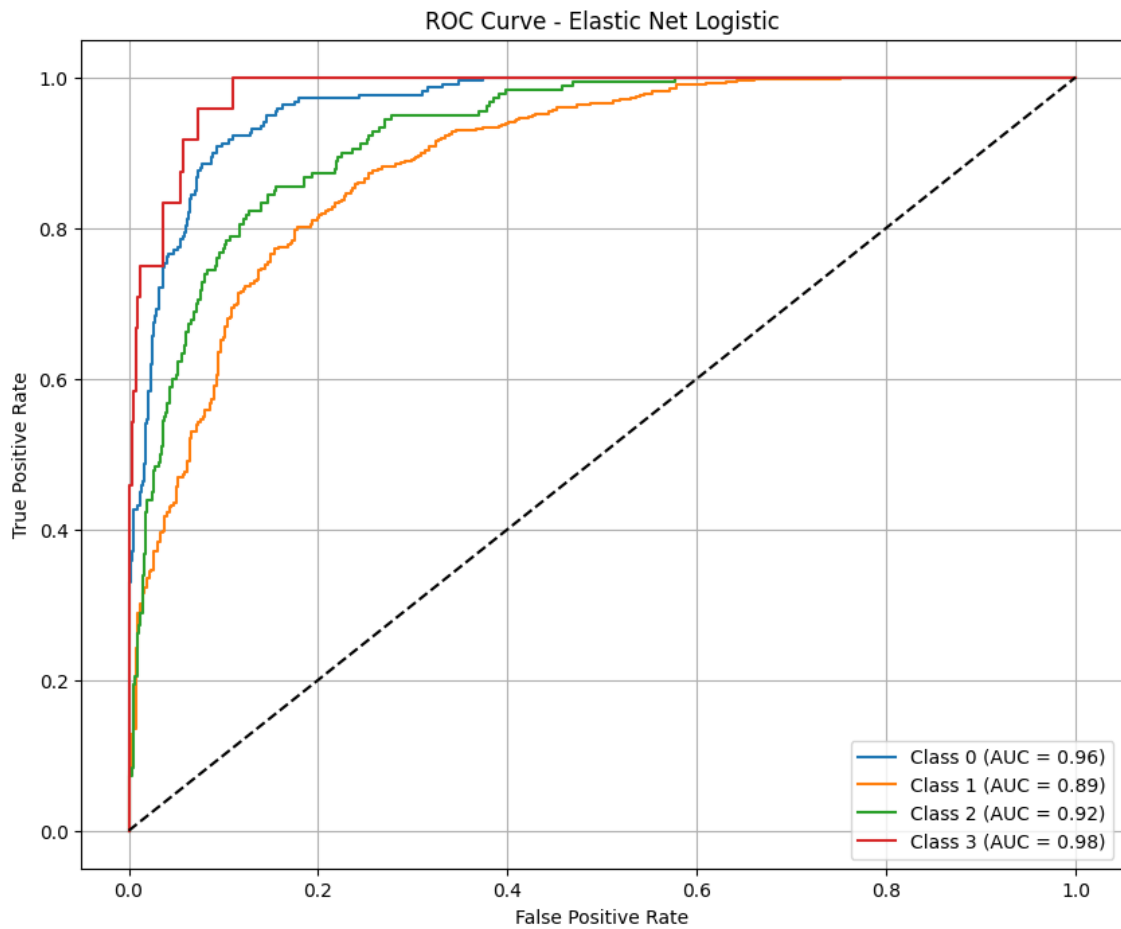


Figure 20: ROC Curve: Elastic Net Logistic Regression

Elastic Net only sets coefficients to zero when the L1 penalty is strong and when the model detects redundant or low-signal parameters. The log-odds plot confirms that coefficients vary in magnitude, but none collapse to zero, with FEAT_SOC1&5 being the strong predictors. Overall, all the ESG features carried meaningful predictive signals, mostly social and environmental.

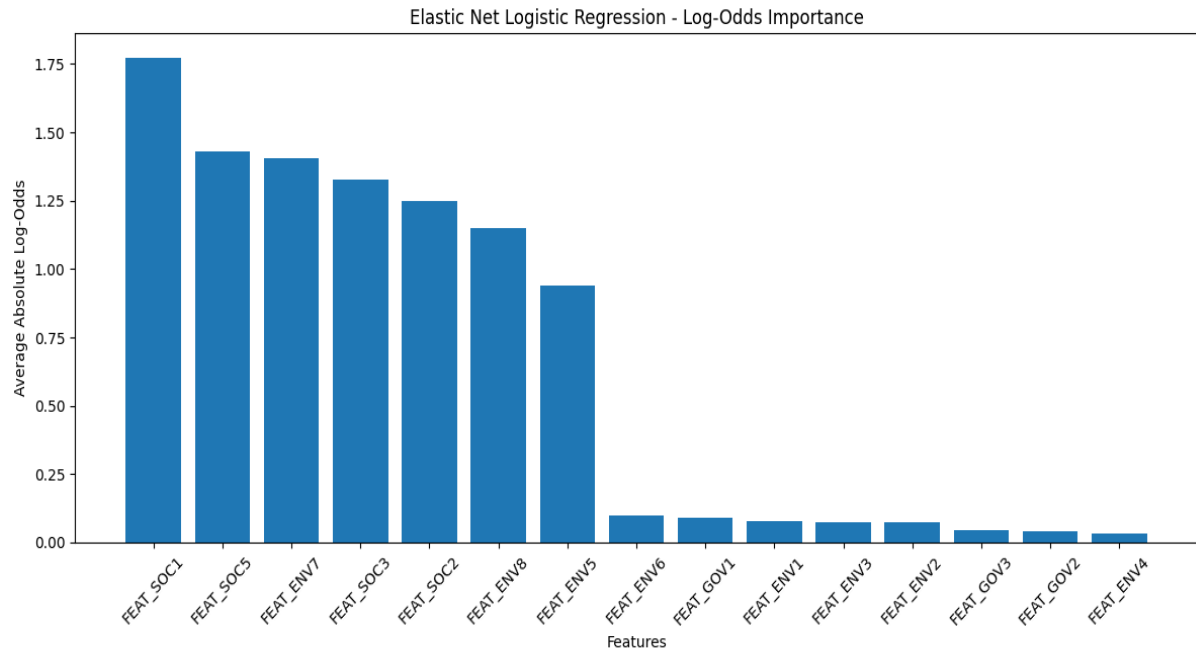


Figure 21: Feature Importance Log-Odds Plot: Elastic Net Logistic Regression

2.2.3. Gradient Boosting

Following the test error rate of $0 \rightarrow 1$ in LR and Elastic net, a stronger tree-based predictive model, Gradient Boosting, is utilised. Gogineni et al. (2026) note that GB iteratively builds smaller models, each learning from the residual errors of the previous one, gradually improving accuracy till the predefined number of iterations is met. This recursive behaviour leads to overfitting and requires careful tuning of parameters such as learning rate and iteration count (Gogineni et al., 2026). Hence, to avoid overly complex learners in interactive boosting, $learning_{rate} = 0.1$, $n_{estimator} = 100$ and $max_{depth} = 3$ is used to keep shallow trees with 0.1 incremental updates, with 100 sequential learners to correct residuals without complexity.

The model showed strong classification performance, with consistently high AUC values across the classes (0.94-0.98), with near-perfect classification on the training set. On the test set, the model struggles to identify ESG Champions (6/24), while performing significantly better than the baseline model in classifying Normal and Winner, given their conceptual proximity. However

$$Test\ Error\ Rate_{0 \rightarrow 1} = \frac{50}{160 + 50} = 22.9\%$$

indicating model weakness of classifying ESG losers as Normal, due to the class imbalance (see Table 5). Although GB generalised well, it poses a higher-risk modelling choice which requires strong control and supervisory expectations (Gogineni et al., 2026). The baseline model performed better in classifying economically meaningful classification errors than GB.

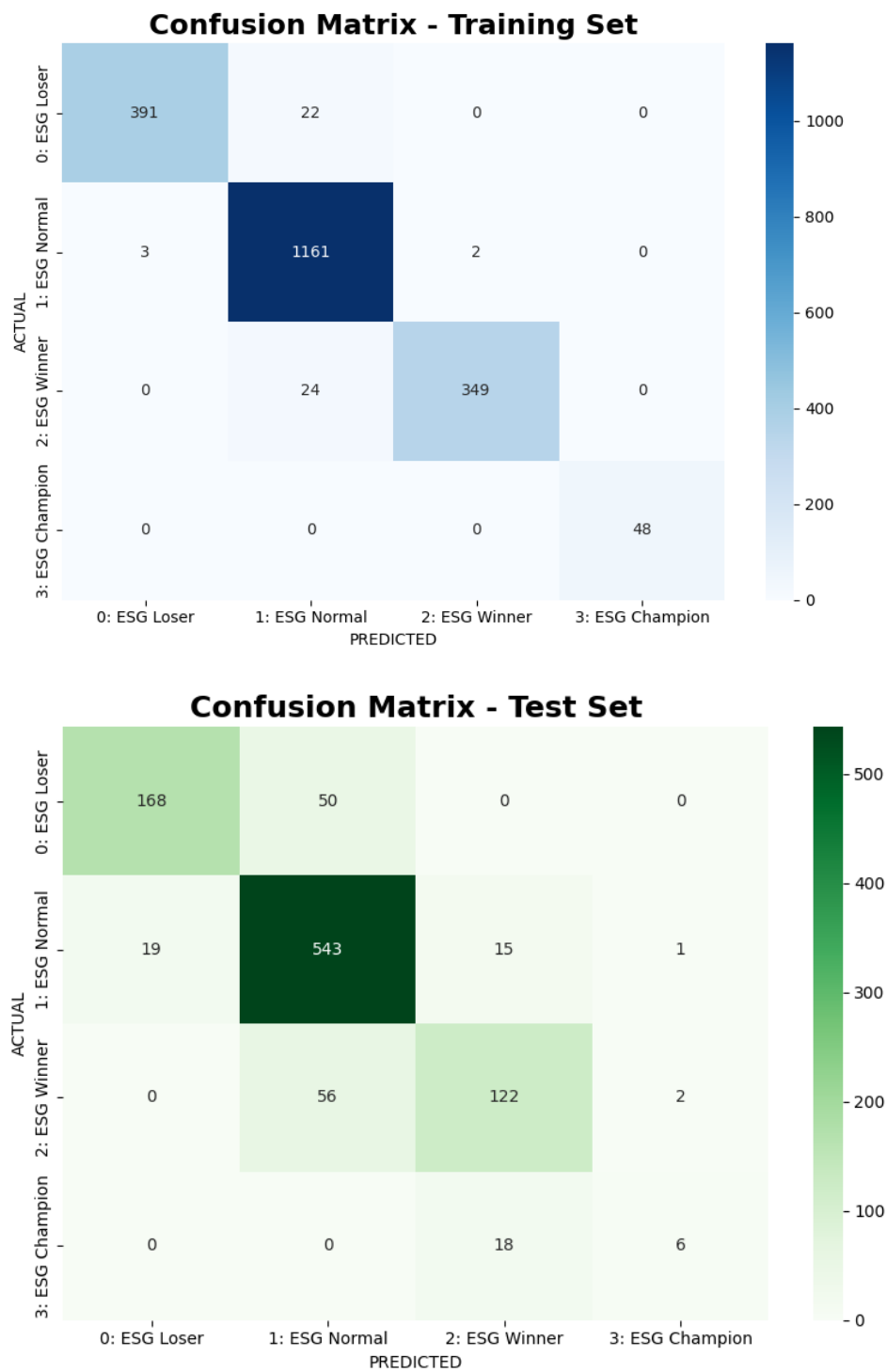


Figure 22: Confusion Matrix on Training and Test Set: Gradient Boosting Model

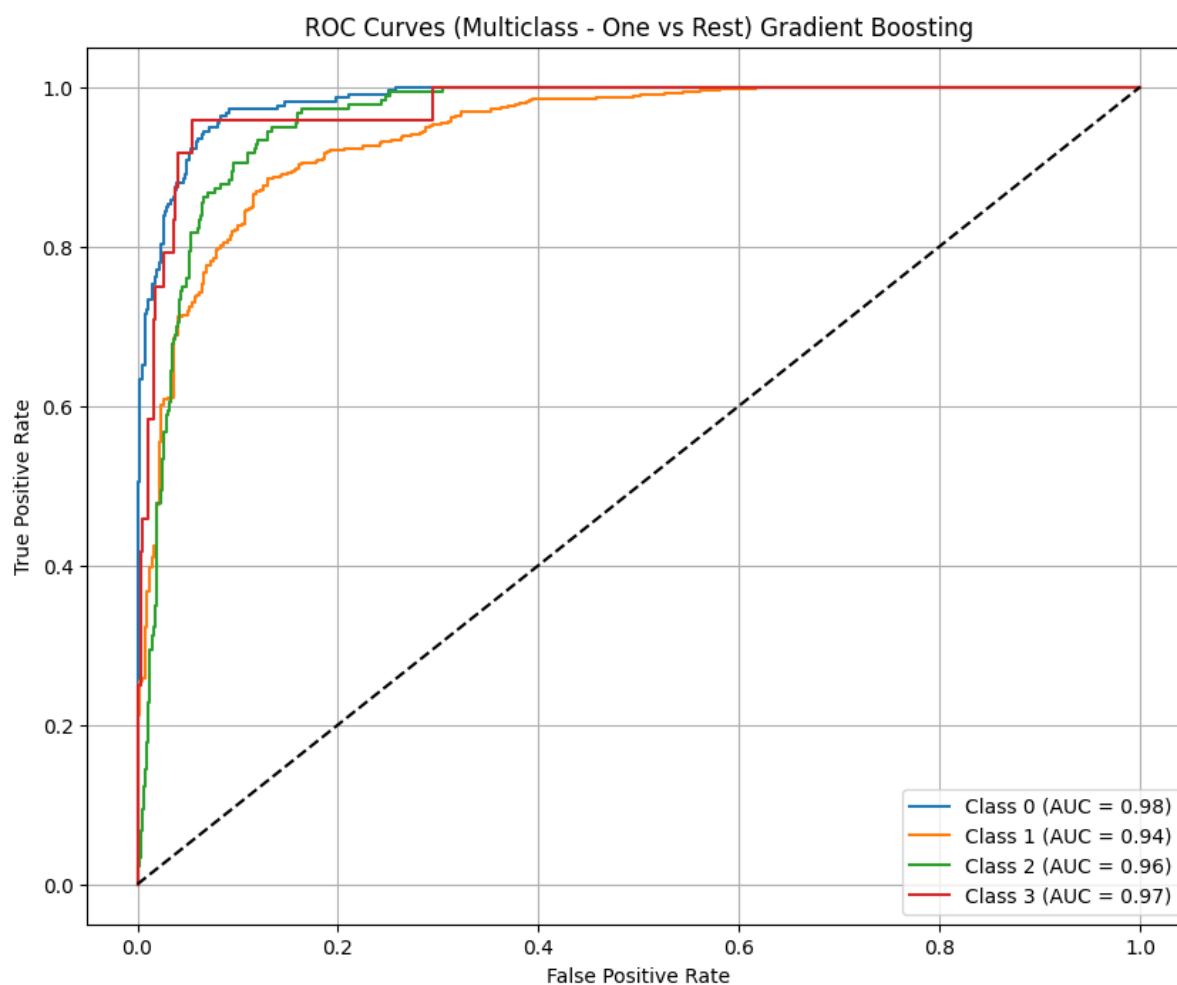


Figure 23: ROC Curves: Gradient Boosting

Furthermore, the feature importance plot in Figure 24 shows that GB weights on social indicators FEAT_SOC1-5 most, while environmental and governance features contribute more diffusely, indicating ESG class separation is primarily socially driven.

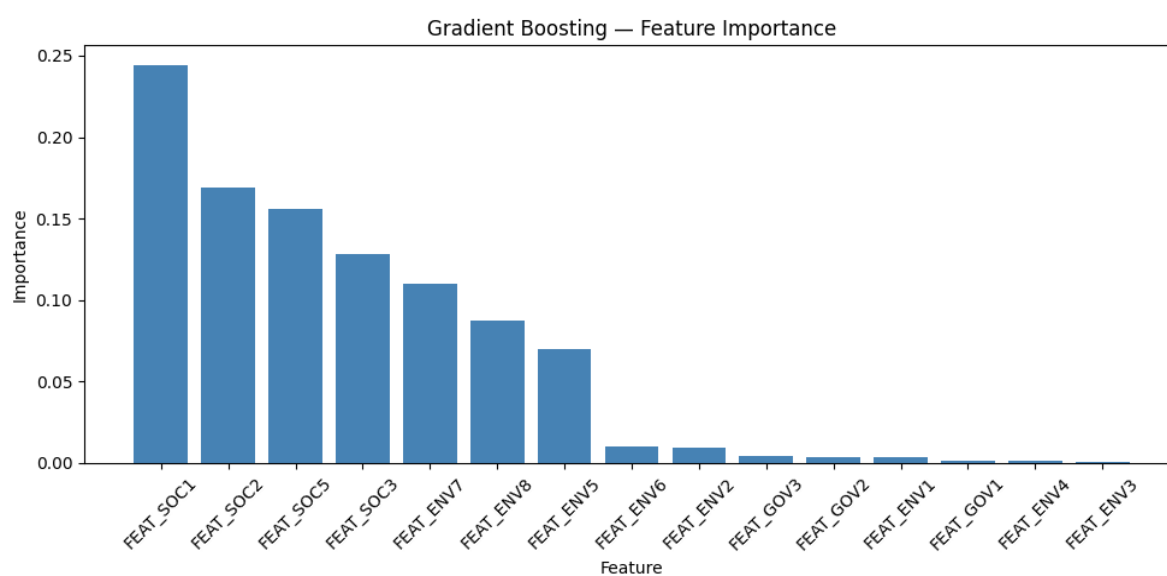


Figure 24: Feature Importance: Gradient Boosting

2.2.4. Random Forest

Random Forest offers a stable, variance-reducing alternative to linear models, using parallel tree aggregation to capture nonlinear ESG relationships (Gogineni et al., 2026). Its predictive accuracy exceeds the baseline model, but regulators view RF as high-complexity, low-transparency, requiring rigorous validation, explainability, and monitoring (Batanova et al., 2023). To address the risk of overfitting and strengthen governance controls, the maximum depth of each tree is capped at 5 to have conservative growth and improved generalisation. The confusion matrices and AUC scores (0.87-0.99) show that RF performs well on the training set but fails to generalise the rare category, ESG Champion, but produces better classification boundaries between Normal and Winner than the baseline model. Moreover, the key economic misclassification between Loser and Normal

$$Test\ Error\ Rate_{0 \rightarrow 1} = \frac{116}{102 + 116} = 53.2\%$$

is not reliably identified, which distorts ESG assessment and requires governance oversight.

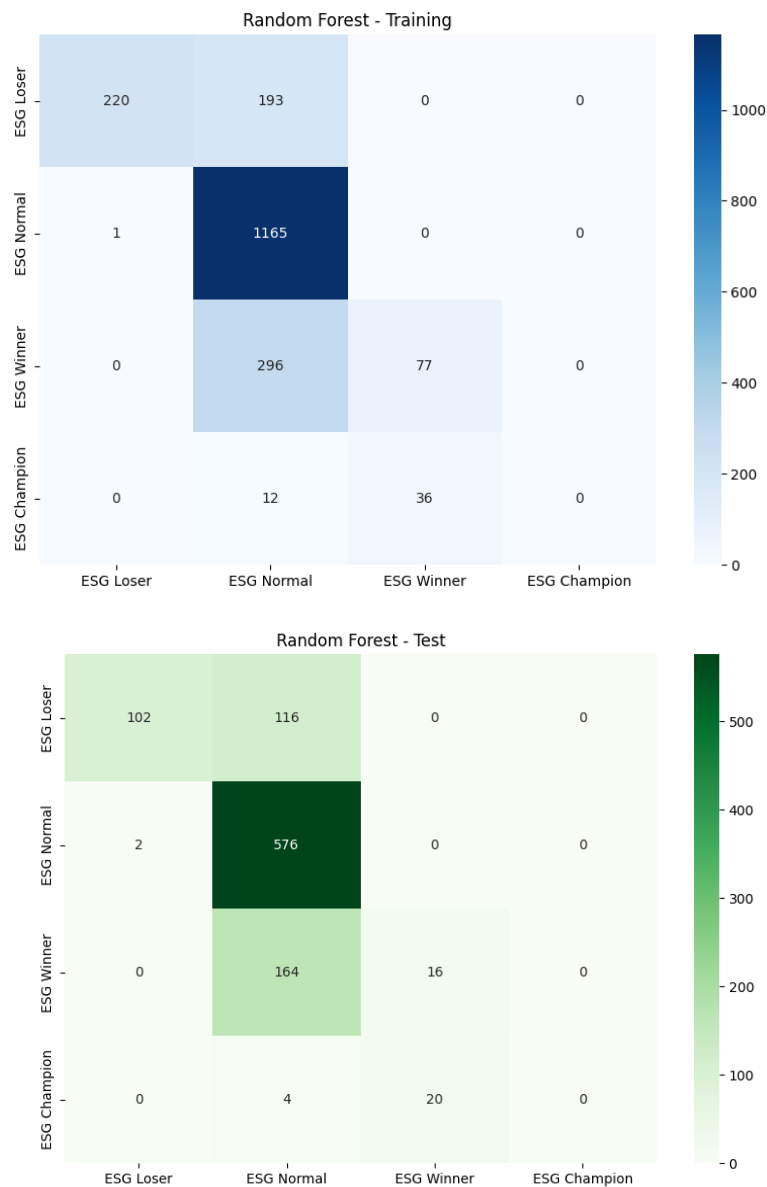


Figure 25: Confusion Matrix on Training and Test Set: Random Forest

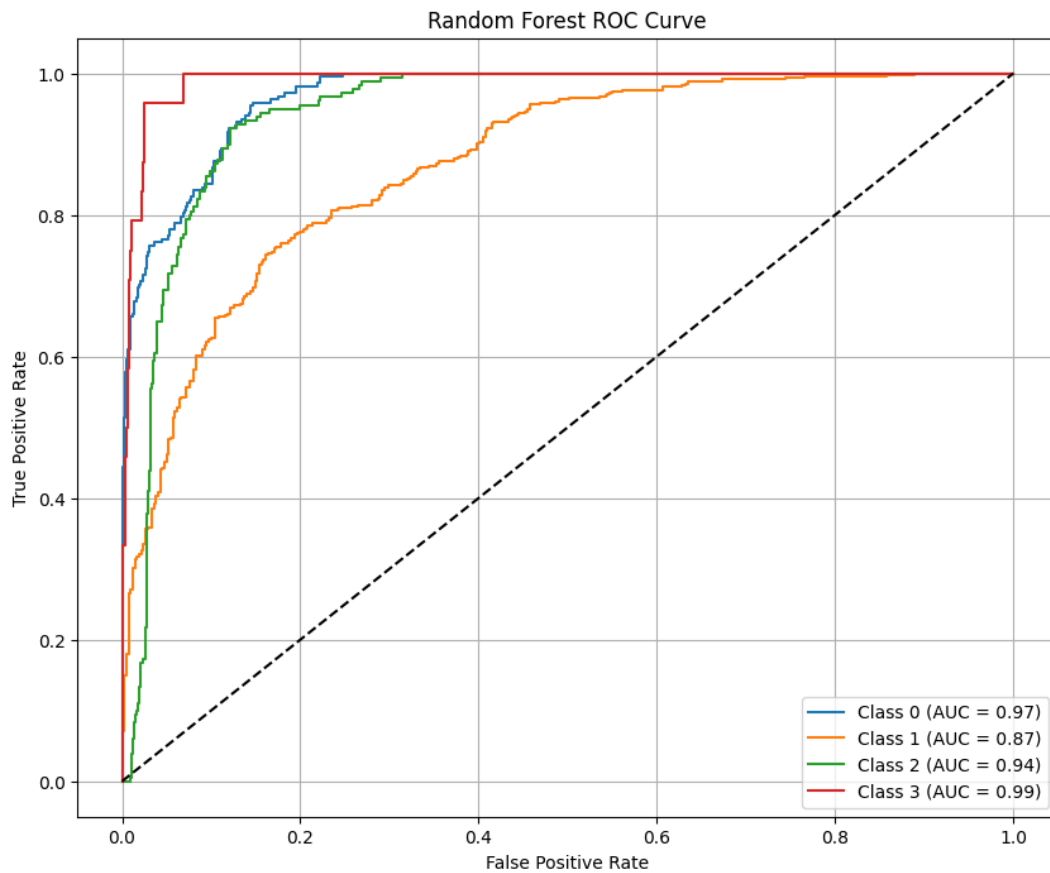


Figure 26: ROC Curves: Random Forest

As illustrated through the above feature importance plots, Random Forest prediction is also derived from social features, with FEAT_SOC1 being the highest predictor. From a governance perspective, the concentration of importance of a few features and the near-zero contribution of governance and environmental features highlight the need for monitoring.

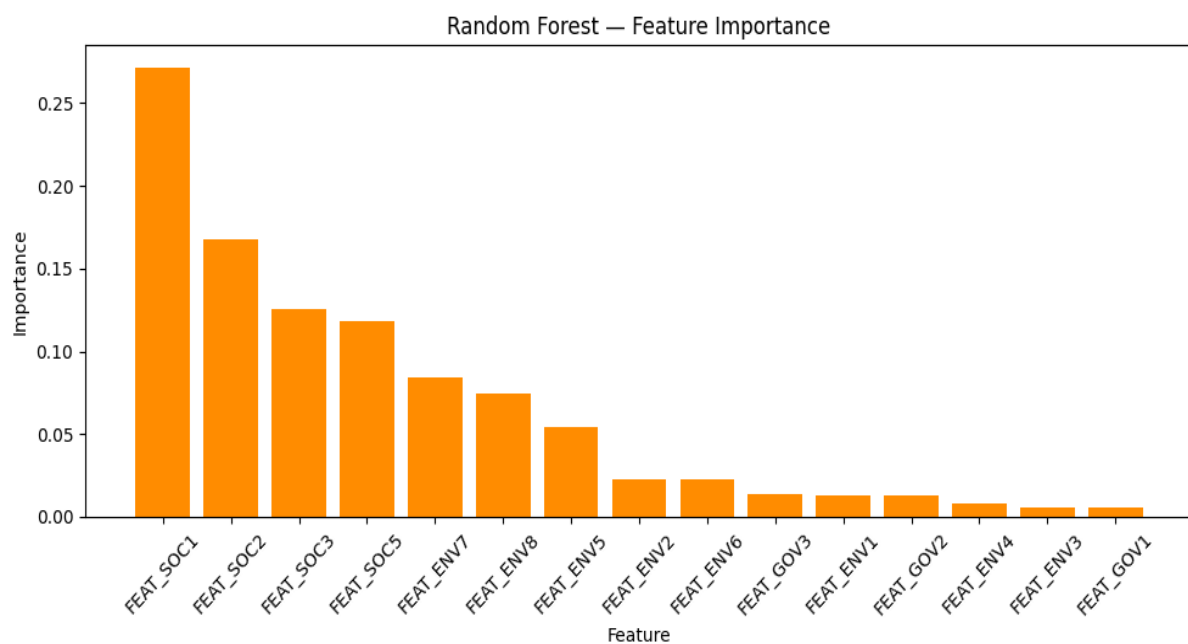


Figure 27: Feature Importance Plot: Random Forest

2.3. Model Comparison

The comparative evaluation of the four fitted models highlights that accuracy alone is not a sufficient metric for the ESG-profiling task. Figure 28 shows that Gradient Boosting achieved the highest test accuracy (84%), followed by Elastic Net (80%), Logistic Regression (74%), and Random Forest performed weakest (69%). From a regulatory perspective, FCA (2019) highlights that although tree-based models are mostly commonly used for predictive performance, they lack explainability and increase model risk, making validation and accountability more difficult. Therefore, simpler and stable models like Elastic Net and Logistic Regression are preferred.

However, on examining the misclassifications in Table 7, predictive accuracy isn't enough of a metric to choose the model. Logistic Regression produces the lowest number of economically harmful errors (20 cases), upgrading ESG Loser to ESG Normal. This has consequences as it artificially inflates the sustainability profile and embeds downside risk into client portfolios. If these errors propagate into live trading strategies, they could lead to misallocated capital, under-priced risk and potentially “greenwashing” (Kathan et al., 2025). ‘Greenwashing’ accusations have high reputational risk, damaging credibility and undermining the integrity of sustainability ratings. Gradient Boosting misclassifies (50), Elastic Net (58), and Random Forest has severe distortions (116), making Logistic Regression the most conservative and governance-aligned model.

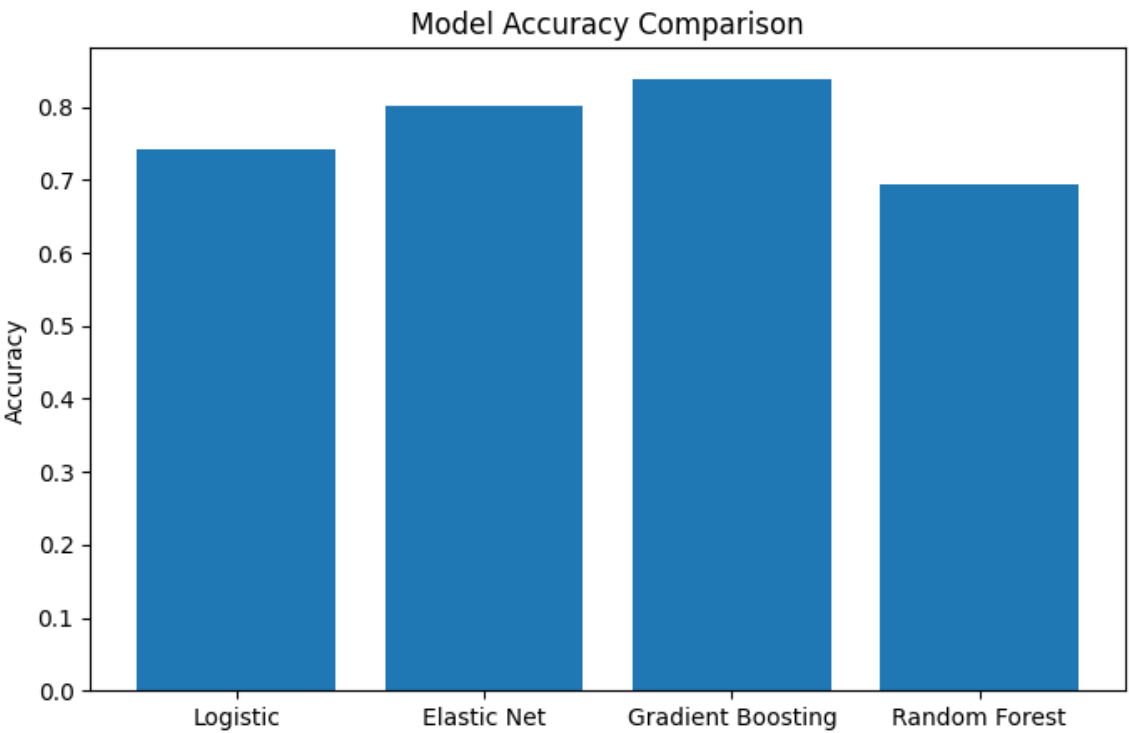


Figure 28: Test Accuracy Scores: Four Fitted Models Comparison

Table 6: Fitted Models Test Accuracy Score

Fitted Model	Test Accuracy Score
Logistic Regression – Baseline Model	74%
Elastic Net Logistic Regression	80%
Gradient Boosting	84%
Random Forest	69%

Table 7: Misclassification Errors across the fitted models

Logistic Regression

<i>Actual Label</i>	<i>Predicted Label</i>	<i>Count</i>
ESG Normal	ESG Winner	103
ESG Normal	ESG Loser	73
ESG Winner	ESG Normal	29
ESG Winner	ESG Champion	26
ESG Loser	ESG Normal	20
ESG Champion	ESG Winner	5
ESG Normal	ESG Champion	1
ESG Winner	ESG Loser	1

Elastic Net Logistic Regression

<i>Actual Label</i>	<i>Predicted Label</i>	<i>Count</i>
ESG Loser	ESG Normal	58
ESG Winner	ESG Normal	58
ESG Normal	ESG Winner	40
ESG Normal	ESG Loser	28
ESG Champion	ESG Winner	12
ESG Winner	ESG Champion	3

Gradient Boosting

<i>Actual Label</i>	<i>Predicted Label</i>	<i>Count</i>
ESG Winner	ESG Normal	56
ESG Loser	ESG Normal	50
ESG Normal	ESG Loser	19
ESG Champion	ESG Winner	18
ESG Normal	ESG Winner	15
ESG Winner	ESG Champion	2
ESG Normal	ESG Champion	1

Random Forest

<i>Actual Label</i>	<i>Predicted Label</i>	<i>Count</i>
ESG Winner	ESG Normal	164
ESG Loser	ESG Normal	116
ESG Champion	ESG Winner	20
ESG Champion	ESG Normal	4
ESG Normal	ESG Loser	2

2.4. Sensitivity Analysis

Table 8: Sensitivity Analysis

Model	Top Feature	Full Accuracy	Reduced Accuracy	Accuracy Drop
Logistic Regression	FEAT_SOC1	0.742	0.536	0.206
Elastic Net	FEAT_SOC1	0.801	0.637	0.164
Gradient Boosting	FEAT_SOC1	0.839	0.680	0.159
Random Forest	FEAT_SOC1	0.694	0.677	0.017

The sensitivity analysis evaluates the dependence of the most influential feature ‘FEAT_SOC1’ on each model. This feature, as identified earlier, has the strongest predictive power. The removal of FEAT_SOC1 results in 20.6% accuracy drop for Logistic Regression, confirming the model relies heavily on this predictor. Elastic Net and Gradient Boosting also show a substantial accuracy drop (16% & 15%), indicating SOC1 is structurally embedded in ESG classification. However, Random Forest shows a minimal accuracy drop (1.7%), suggesting that although SOC1 is important, the model distributes predictive weight across many variables. From an investment portfolio perspective, sensitivity analysis highlights that SOC1 is the dominant ESG driver; any instability or bias in this feature would propagate through all models. Although Logistic Regression is better at generalising with new data, its dominance on SOC1 raises concerns about model stability if any shifts or distortions occur in this single feature.

2.5. Cross-Validation

K-fold cross-validation is a model validation technique that tests how reliably a machine-learning model will perform on new, unseen data by repeatedly training and evaluating it on different slices of the data (T R. et al., 2023). The test dataset was split into 10 folds, training the model on $k - 1$ folds and testing on the remaining fold. This process repeats itself until every fold has acted as the test set once, and performance scores are averaged.

The same models are fitted again using cross-validation. Logistic regression and Elastic Net exhibit a tight spread with consistent performance on the training and test sets, confirming their regulatory suitability (FCA, 2019), and low variance to sampling variation. Gradient boosting shows a substantial gap between the training and test sets, showing overfitting. Random Forest shows the weakest generalisation and lowest cross-validated accuracy. F1-Macro measures the model’s average classification quality across all ESG categories (Gogineni et al., 2026). Elastic Net performs best, with balanced accuracy across all ESG classes.

Table 9: Model Accuracy summary across fitted models after using Cross Validation

Model	Train Accuracy Mean	Train Accuracy Std	Test Accuracy Mean	Test Accuracy Std	F1 Macro Mean	F1 Macro Std
Logistic Regression	0.7548	0.0044	0.7425	0.0282	0.6843	0.0547
Elastic Net	0.8147	0.0034	0.8050	0.0249	0.7095	0.0754
Gradient Boosting	0.9790	0.0025	0.8560	0.0249	0.6868	0.0736
Random Forest	0.7259	0.0062	0.6800	0.0219	0.3764	0.0308

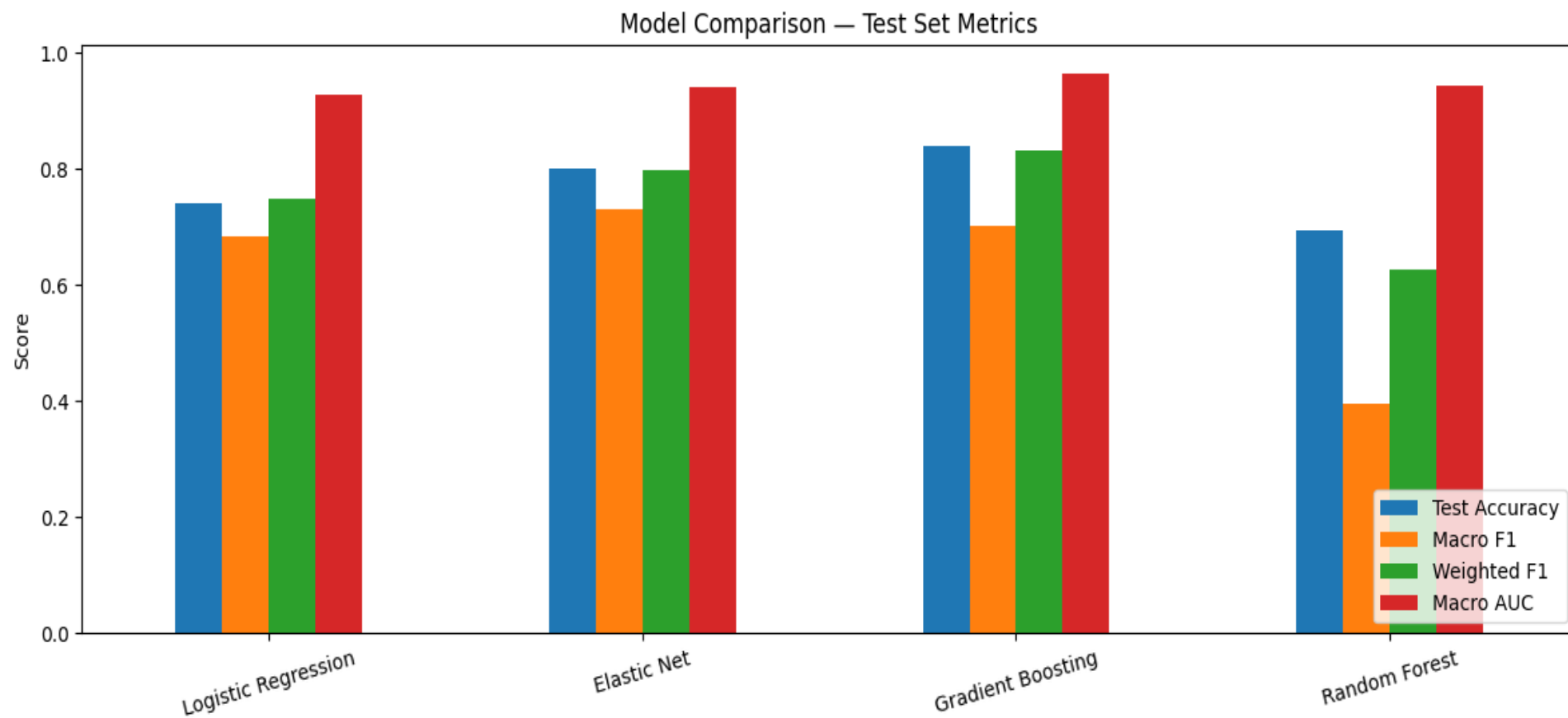


Figure 29: Model Comparison Using 10-Fold Cross Validation: Across Test Accuracy, Macro F1, Weighted F1 and Macro AUC

Following the analysis, Logistic Regression produced the lowest major test misclassification error between ESG Loser and ESG Normal, with Gradient Boosting performing second-best. Elastic Net achieved the highest mean F1-Macro (0.71) after cross-validation, indicating a good balance across all classes, but it did not minimise the most critical misclassification error. Random Forest performed the weakest on the test set.

2.6. Classification Boundary Analysis using PCA

This section focuses the analysis on the model behaviour of Logistic Regression and Gradient Boosting. As the above sections clearly indicate, Logistic Regression is the most stable, with a high-test accuracy score and a wide acceptance from regulators. On the other hand, Gradient Boosting achieved the highest test accuracy with a similar F1-Macro score to LR, indicating that it can also handle class imbalance well. The purpose of this section is to understand whether a linear classification boundary as produced by Logistic Regression is sufficient for separating ESG categories or whether the more flexible, non-linear boundaries generated by Gradient Boosting provide a meaningful improvement.

PCA is used to project high-dimensional ESG feature space into two principal components to visualise how well the models separate the four ESG categories. Firstly, the training dataset was transformed into a new set of uncorrelated variables, the principal components, which capture the greatest variance in the original data. This approach is not used to measure performance, but to understand model behaviours, interpretability, and the stability of the decision boundaries.

i. Logistic Regression

Figure 30 shows that Logistic Regression uses a rigid linear decision boundary that partitions ESG categories, with the top left corner concentrated with ESG Loser classification and the bottom-right being the ESG Champion classification. However, Figure 32 reveals a critical observation of the overlapping regions between the ESG Loser and the Normal. Although LR performs better at classifying ESG Loser, with only 9% error rate (20 cases, see Table 7), the rigid linear boundaries misclassify 15% ESG Normal as ESG Loser (73 cases, see Table 7). The ESG Loser to ESG Normal (9%) is the ‘greenwashing risk’, as a genuinely poor performing firm is screened as acceptable for a sustainability -focused client portfolio. On the other hand, ESG Normal to ESG Loser (15%) is a ‘reputational risk’, as it represents an exclusion of an acceptable firm from the portfolio, a missed opportunity to gain returns.

ii. Gradient Boosting

Figure 31 shows that Gradient Boosting uses a non-linear boundary to capture complex ESG interactions and learns well on highlighting the ESG Normal, the dominant class and fails to correctly identify the ESG Champion category. Figure 33 reveals that even a complex nonlinear boundary cannot separate ESG Loser and ESG Normal. It also mostly misclassifies ESG Winner as ESG Normal, 31% (56 cases, see Table 7). This reveals the inherent structure of Gradient Boosting as an ensemble method, overlearns on dominant classes and fails to differentiate between smaller adjacent ESG Loser (20.7%) and ESG Winner classes (18.7%). Subsequently, classifying ESG Normal to ESG loser is only (3%) (19 cases, see Table 7). Hence, Gradient Boosting reverts the ‘reputational risk’ but amplifies the ‘greenwashing risk’ by classifying poor performing firms as acceptable firms, which in live trading poses regulatory concern.

Logistic Regression — Per-Class Decision Boundaries (One-vs-Rest)

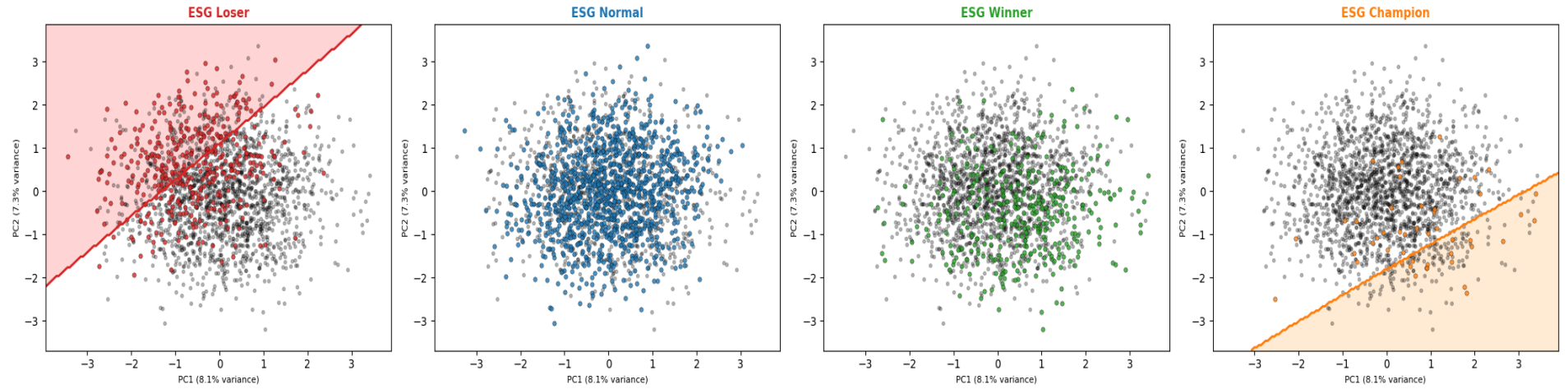


Figure 30: Two-Dimensional PCA Projection of Classification Boundaries: Logistic Regression – Baseline Model

Gradient Boosting — Per-Class Decision Boundaries (One-vs-Rest)

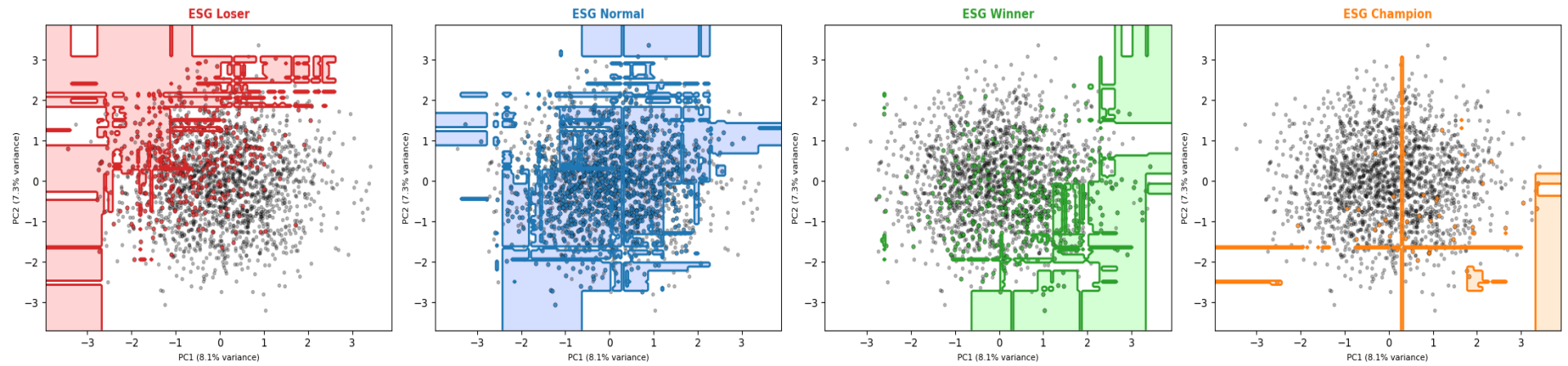


Figure 31: Two-Dimensional PCA Projection of Classification Boundaries: Gradient Boosting

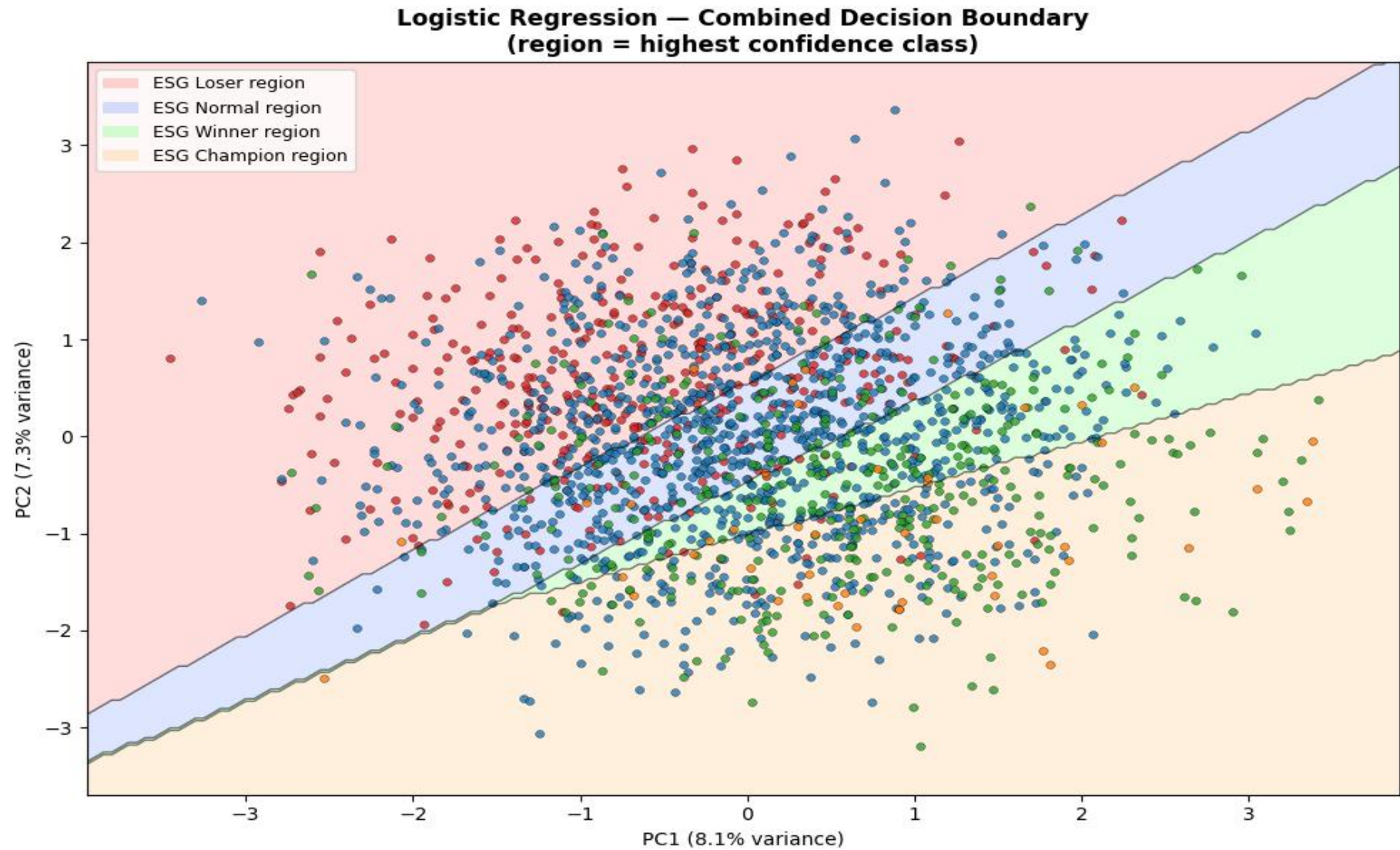


Figure 32: Combined Plot of PCA Projection of Classification Boundaries: Logistic Regression – Baseline Model

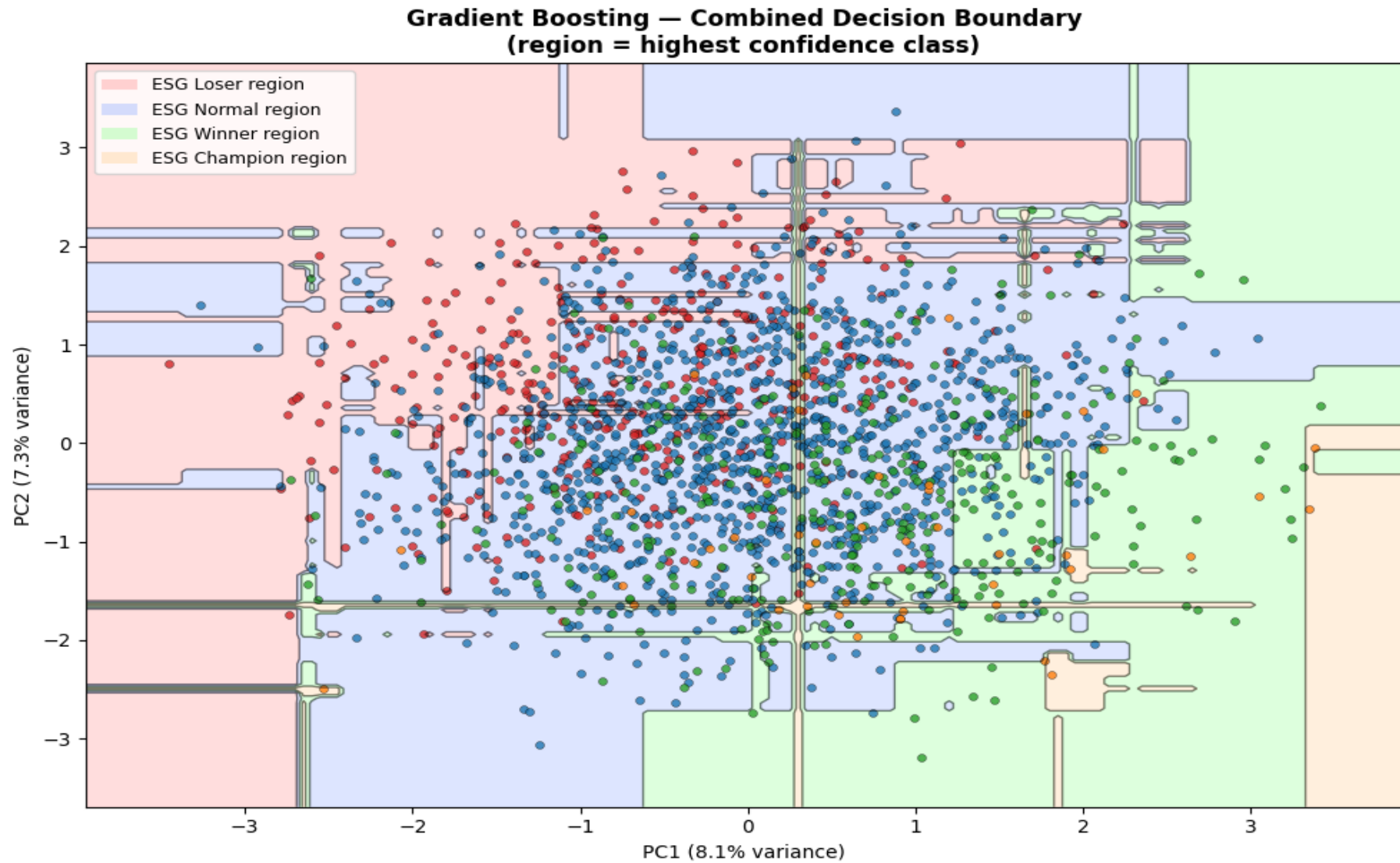


Figure 33: Combined Plot of PCA Projection of Classification Boundaries: Gradient Boosting

Across both models, the PCA analysis confirms that the ESG Loser-Normal boundary is the most structurally ambiguous region in the feature space, and no model fully resolves it. Given the asymmetry between greenwashing risk and reputational risks, Logistic Regression's conservative linear boundary minimises the most economically meaningful error Loser→Normal.

2.7. Recommendation

Logistic Regression is recommended for deployment as the primary ESG classification model. Its interpretability allows compliance and helps document and challenge any classification decision. Under MiFID II RTS 6, any ML or algorithmic model must be fully governed, independently validated, auditable, evidence-based, and subject to annual self-assessment, with clear accountability and robust controls to prevent operational or market harm (Cankett and Farrell, 2019). Hence, from the regulatory perspective, Logistic Regression's explainability and predictive behaviour therefore aligns well with MiFID II RTS 6 governance expectations.

The cross-validation confirms that Logistic Regression is the most stable, with a test accuracy standard deviation of 0.028, and generalises well with new data with a lower risk of overfitting. As the sensitivity analysis highlighted the LR's dependence on SOC1, ongoing monitoring and data quality assurance are required to avoid bias, distortion, and instability in the model's output.

Furthermore, the most sequential error, ESG Loser, was misclassified as ESG Normal, which is handled most well by LR. This error constitutes a 'greenwashing risk' under SFDR Regulation 2019/2088, exposing investment clients to undisclosed ESG risk and subjecting them to regulatory sanctions (European Parliament and Council, 2019). Subsequently, the LR's test error on classifying the ESG Normal as Loser, is 'reputational risk' rather than regulatory, as it doesn't mislead clients but leads to 'lost opportunity' in investment.

Hence, the deployment of Logistic Regression as an ESG classifier should be conditional on periodic revalidation, human oversight of the boundary-zone classifications, and clear documentation of model behaviour to ensure continued compliance with MiFID II RTS 6 and SFDR expectations.

Component 3

The analysis in Components 1 & 2 exposes structural limitations, such as central-tendency bias, heavy single-feature reliance (FEAT_SOC1), and weak ESG indicator features, like governance and environmental. These weaknesses reflect a systematic problem in data quality rather than model instability. This mirrors Berg, Kolbel and Rigobon (2022), who show that 56% of ESG rating divergence arises from measurement inconsistency, with ESG ratings for the same firms aligning weakly (38%). This is due to overreliance on standardised, self-reported corporate disclosures and persuasive ‘rater-effect’ (Berg, Kolbel and Rigobon, 2022), meaning that ESG scores often reflect both limitations of underlying data and rating agencies’ overall impression of the firm rather than objective, independent verified indicators. These limitations suggest improving the data quality for better ESG categorising.

3.1. Use of automation

Digitalisation can address the data integrity and support evidence-based sustainability decision-making through automation, which enables systematic collection, consolidation and cleaning of sustainability metrics (Telukdarie et al., 2026). AI enables the transformation of unstructured alternatives like News data, satellite imagery, social media and temporal dimension into structured ESG signals (Telukdarie et al., 2026). The real-time processing of high-frequency unstructured data, such as news articles, social media, and regulatory updates, in traditional ESG data can speed up the update of new ESG risks or events. This allows models to update ESG categories without manual oversight. Following the analysis in the Component, Logistic Regression fails to identify ESG Loser↔Normal ↔Winner due to the ambiguity in the predictors which help differentiate between Normal and Winner. To strengthen the indicator association, adding news article data, which gives real-time updates on firms' performance for ESG goals, can make the differentiation between categories apparent.

Moreover, AI can enhance coverage by incorporating alternative data sources like satellite images, geo-spatial analytics, sensor data, and web-scraped content (Sustainability Directory, 2026). Traditional ESG data often depends on annual reports, which are widely available for large publicly listed companies or banks. However, for SMEs (small to medium-sized), annual sustainability/ESG reports are not required, which restricts from securing green capital, attract sustainable investments or fully integrate into carbon-conscious corporate supply chains (Ozkan et al., 2023). AI techniques can process alternative data for such companies to generate ESG insights.

AI can also improve the reliability and quality of the ESG signal by reducing the reliance on self-reported data and enabling cross-validation across multiple independent sources. As indicated in 1.5, category labels are produced through subject-matter expert judgement without a documented scoring methodology, subject to bias, inconsistency and greenwashing (Telukdarie et al., 2026), as firms may selectively disclose favourable information. Using NLP-based methods, the inconsistency between company discourses and external narrative can be identified to improve the overall reliability and quality of ESG metrics (Davidescu et al., 2026).

3.2. Unstructured Data Sources:

News and media sentiment pipeline by LLMs to read corporate reports, environmental incidents, governance failures, labour disputes, regulatory actions and stakeholder sentiment (Kou et al., 2025). It can synthesise thousands of articles, NGO reports, regulatory bulletins, and social media narratives and generate a dynamic, forward-looking view of a firm's ESG risk profile. This can improve the model's dependency on a few associated indicators, like social factors, dominantly in the fitted models, to a wide range of diverse inputs. This would decrease the bias or error propagating in the model if the strongest indicator structural distortion. Moreover, it can also improve the 'greenwashing risk' identified from the fitted models, as real-time controversy monitoring through LLMs can improve the identification of ESG Losers passed through the models, making it suitable for live trading strategies.

Furthermore, environmental components showed minimal association in ESG categories. This can be improved by using satellite imagery. High-resolution satellite sensors can detect methane plumes, track deforestation, monitor land-use change, and measure air and water quality at scale, offering independent verification of firms' environmental claims (Gu et al, 2023). This enables early detection of anomalies or harmful activities that annual ESG reports often miss. This also strengthens the verification of environmental performance evidence across regions.

LLM-based textual analysis, information extraction, and satellite-derived environmental signals can improve ESG screening and enhance the accuracy of portfolio construction.

3.3. Regulatory and Long-term risk oversight on AI adoptions

Under MiFID II RTS 6, any AI-automated ESG classification needs to be audited, documented, and independently validated (Canktt and Farrel, 2019). The EU AI Act (European Commission, 2024) further classifies ESG-relevant AI models as high-risk, requiring transparency, robustness testing, human oversight and post-deployment. This is particularly important for the LLM-based real-time controversy pipeline. Hallucinations. Distorted or false sentiment flags can amplify bias and misinformation. Moreover, as LLMs synthesise news, reports and social media explainability control, including. Satellite-based environmental verification introduces additional regulatory expectations. Under SFDR (European Parliament and Council, 2019), environmental claims must be independently verifiable; therefore, methane detection or deforestation requires documentation, accuracy thresholds and cross-validation against firm-reported emissions or other third-party datasets for anti-greenwashing requirements.

The long-term risk of AI adoption is model drift. This means that AI systems calibrated on historical ESG data risk systematics misclassification, generating signals that are structurally backwards-looking relative to the regulatory frontier (FSB, 2017). This is also noted by Kotsnatonis and Serefeim (2019) as a fundamental data architecture error, where ESG metrics lack a standardised definition across jurisdictions, meaning AI models trained on heterogeneous inputs will inherit and amplify pre-existing measurement inconsistencies at scale. Avramov et al. (2022) further add that ESG rating "uncertainty constitutes not just idiosyncratic noise but a systematic factor" propagating material into the portfolio constructions. In automated pipelines, this means that small inconsistencies in training data can develop as persistent misclassification errors unless the models are continuously recalibrated and monitored. Strategically, the adoption of LLM-based controversy monitoring and satellite-derived environmental verification can generate an information advantage in strengthening environmental and governance signals, reducing the dependence on SOC1-driven classification.

In conclusion, the structural limitation identified across Components 1 & 2 reflects systemic ESG data quality failure rather than model error. AI-driven integration can improve ESG features coverage, reliability and classification accuracy. However, the sustained development demands strict regulatory compliance with MiFID II RTS 6 and SFDR for continuous model recalibration, rigorous explanatory controls and documentation for annual audits.

Use of Gen AI Statement:

I acknowledge the use of Claude Sonnet 4.6 (Anthropic, <https://claude.ai/>), CoPilot 365(Microsoft, <https://copilot.microsoft.com/>) and VS Code(Microsoft, <https://code.visualstudio.com/>).

I used Claude Sonnet 4.6 to get an outline for Component 2, understanding key areas to address in the analysis using coding. After the coding part, I used Claude to figure out discrepancies in my code, like variable shadowing, as I kept reusing the variable 'model' again in storing the fitted models and evaluating predictions. This helped me write clear code syntax without worrying about producing wrong results. Furthermore, Claude assisted in sensitivity analysis, as I was confused about how to understand the feature selection impact on the models. The PCA boundary section was also assisted by Claude. I attached the Lecture 5 PDF and was prompted to suggest code which visualises the decision boundary figure as illustrated in the notes.

I used Copilot 365 to summarise the articles, especially the European Parliament and Council Article for detailed SFDR regulations and their impact. I also used CoPilot to paraphrase my technical Component 1 analysis to the wealth manager audience. This helped to work on the formative feedback effectively, as CoPilot generated a good draft, but it lacked a clear link with references from external sources and read like a superficial opinion. Hence, I had to read and highlight the external sources to have a clear understanding of the analysis and prevent hallucinations in the explanations. Moreover, CoPilot also assisted in rechecking that my analysis is sound and coherent with the audience, like in explaining key ML techniques, I would prompt CoPilot to explain to a non-technical audience. This helped me avoid too much technical jargon and ensured my writing remained accessible.

I used VS Code for the coding. The AI assistance was used in Data Leakage on creating the table of association with respective features and categories in descending order. I also prompted VS Code AI assistant to generate separate correlation matrices for Low-Ordinal and Pearson's for High-ordinality and numeric features. Additionally, I used it to troubleshoot minor syntax issues, improve readability, and ensure consistent formatting across the notebook. While the AI tools supported my workflow, all analytical decisions, interpretations, and conclusions were made independently, based on my understanding of the module content and external readings.

Out of curiosity, I used Claude Sonnet 4.6 on Max effort to mark my assignment using the Assignment Brief, rubric and VS Code File. Although the marking seemed optimistic, it was still useful as a sense-check to identify areas where my explanations lacked clarity or where the justification for model choice needed strengthening. This helped me revisit those areas with a more critical lens. Importantly, I did not rely on the AI's grade prediction but instead used its comments to reflect on whether my arguments were sufficiently evidenced, coherent, and aligned with the learning outcomes. This prompt utilises a lot of tokens and runs out of free chats instantly after the response, so only 1 attempt was made, which was entirely sufficient for my needs.

Word Count: 5485
Component 1: 1481
Component 2: 2504
Component 3: 1000
AI Statement: 500

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